

Dark-matter-independent orbital decay bounds on ultralight bosons from OJ287

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Based on 2505.09696, 2510.27424

In collaboration with

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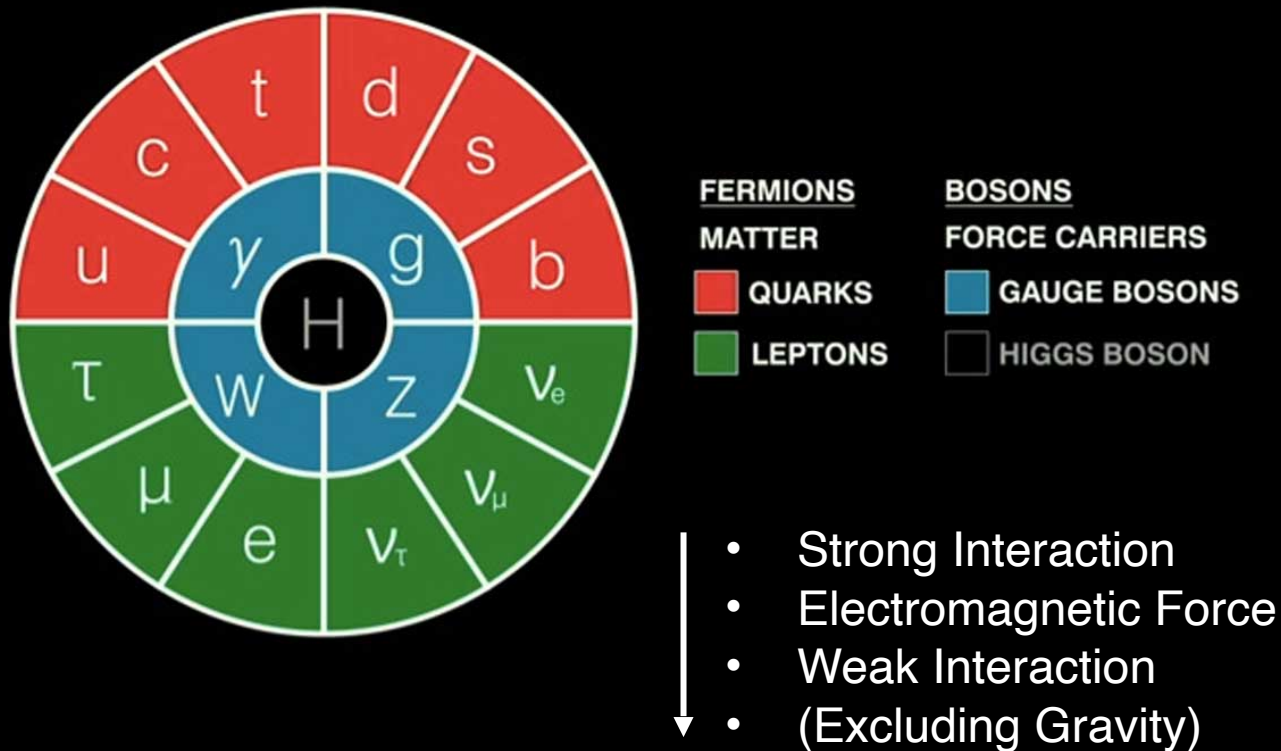
Minxi He (IBS CTPU-PTC)

Volodymyr Takhistov (QUP, KEK)

Outline

- Introduction of Black Hole Superradiance
- Gravitational Atom Phenomena
- Application on OJ287
- Other Application...

Standard Model?



Unsolved mysteries Beyond Standard Model (BSM)

- Galaxy rotation curves
- Galaxy clusters
- Gravitational lensing
- ...

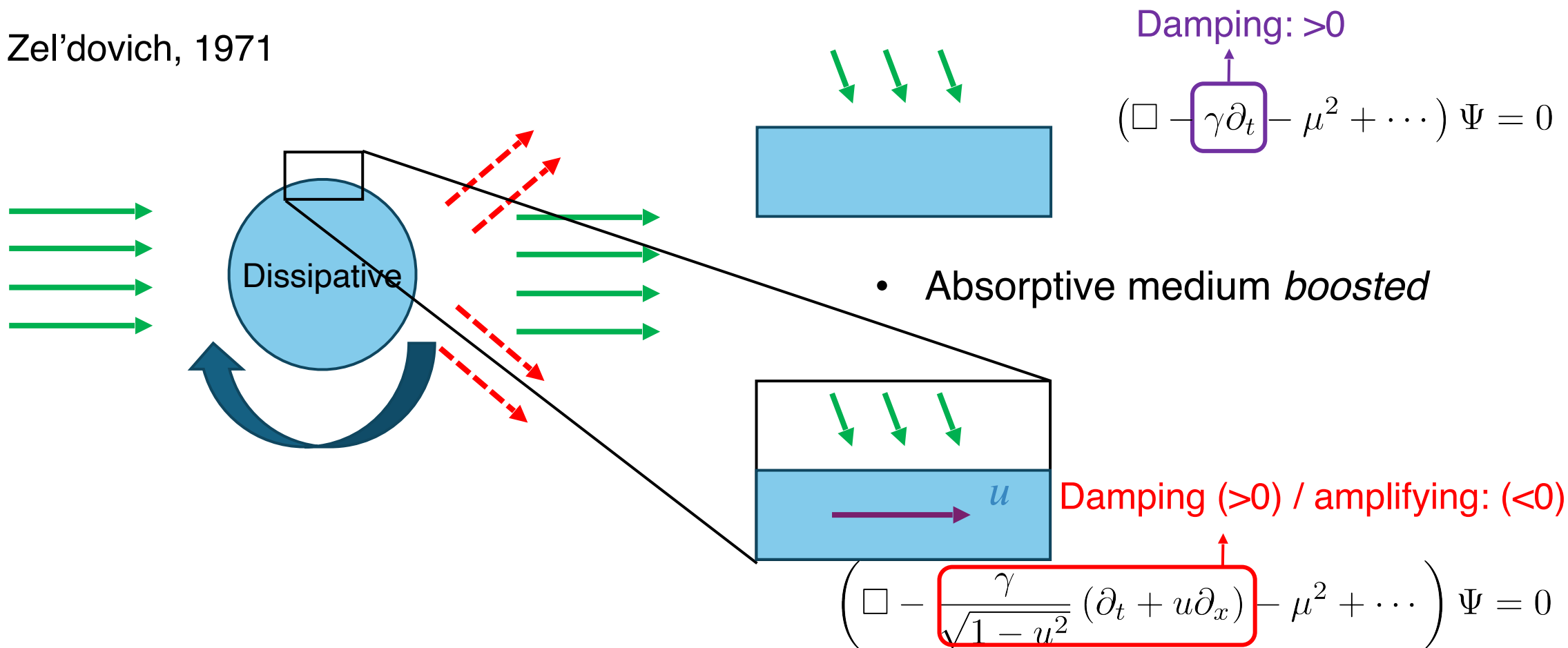
Colliders?



BH is a good tool to test BSM Particles

Rotational superradiance • Absorptive medium *at rest*

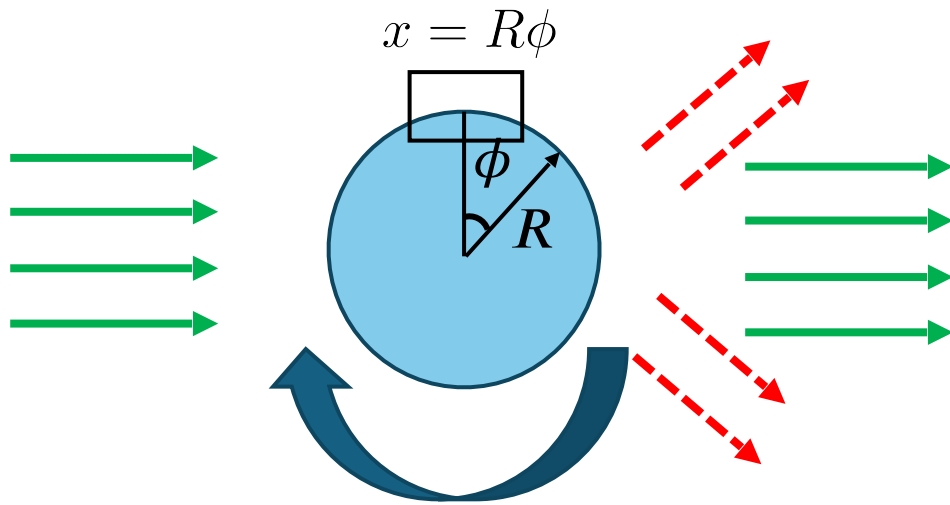
- Zel'dovich, 1971



Lesson: “Slow” modes with $-\frac{\partial_t}{\partial_x} \sim v_{\text{ph}} < u$ are actually **amplified** instead of absorbed

Rotational superradiance

Lesson: “Slow” modes with $-\frac{\partial_t}{\partial_x} \sim v_{\text{ph}} < u$ are actually **amplified** instead of absorbed



“Slow” in the rotational context:

$$\Psi \sim e^{-i(\omega t - m\phi)}$$

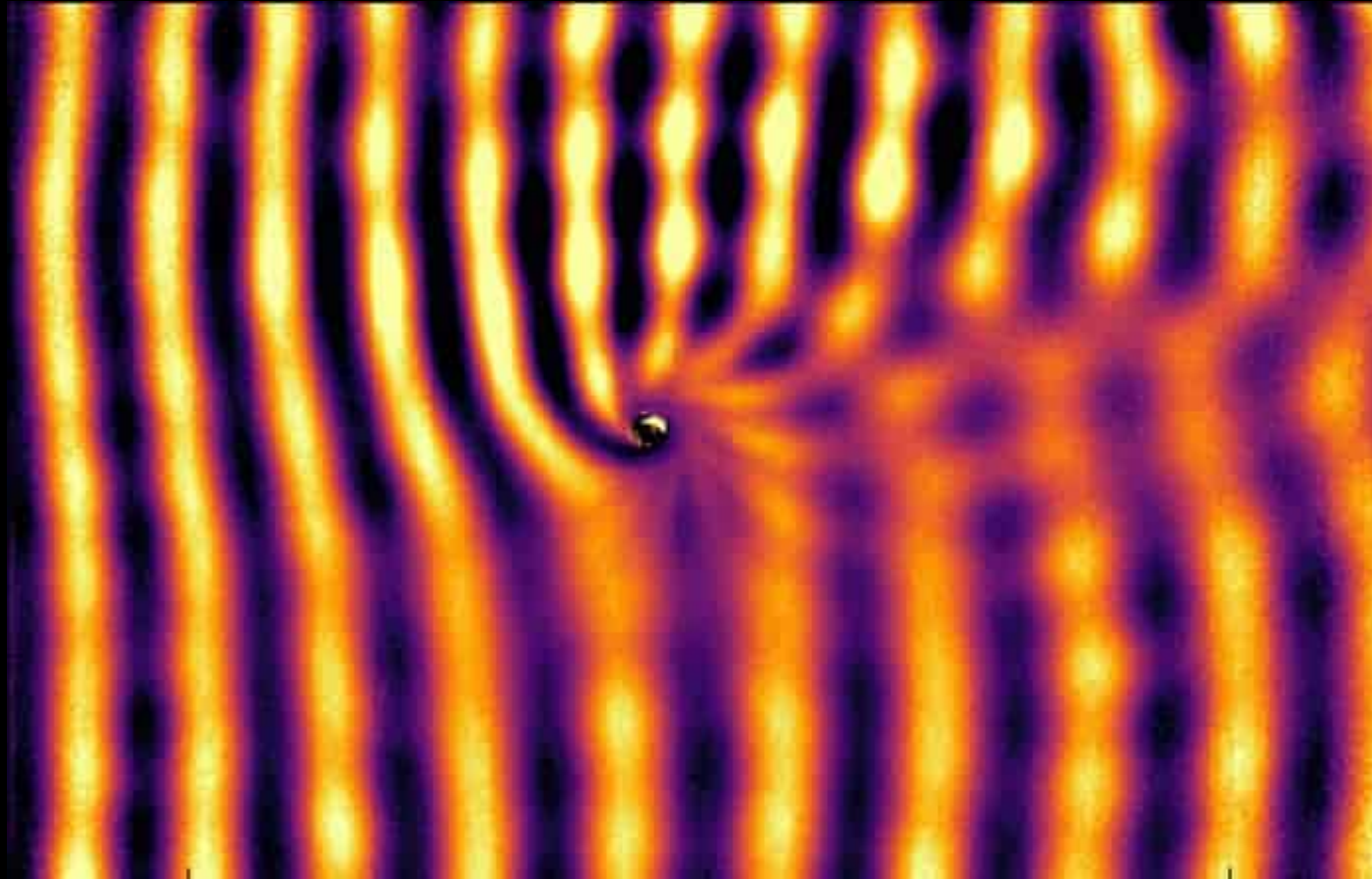


$$-\frac{\partial_t}{\partial_x} \sim \frac{\omega}{\partial_{(R\phi)}} \sim \frac{\omega}{m/R} < u \sim R\Omega$$



Amplification criterion:

$$m\Omega > \omega$$



Water wave superradiance by a vortex drain [Torres et al., 2016]
(20% amplification)

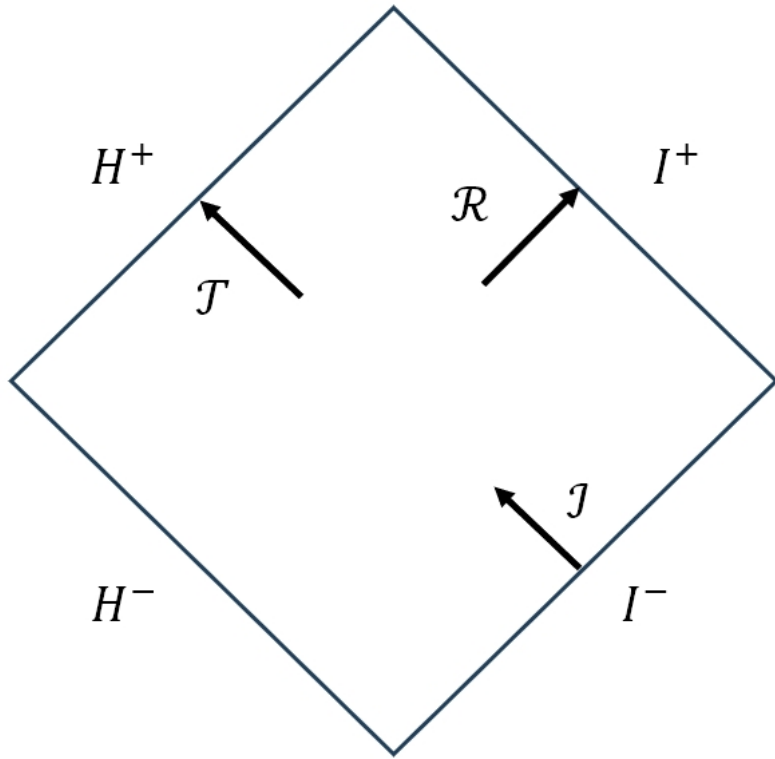

$$m\Omega_H > \omega \sim \mu$$

Same thing for a black hole and a boson

[Press & Teukolsky,
1972]

[Damour et al. 1976]

Superradiant Scattering



KG Equation in Kerr with tortoise coordinate:

$$dr_* = \frac{r^2 + a^2}{\Delta} dr ,$$

Asymptotic behavior:

$$\chi = \begin{cases} e^{-i\omega r_*} + \mathcal{R}e^{i\omega r_*} , & r \rightarrow \infty ; \\ \mathcal{T}e^{-ik_H r_*} , & r \rightarrow r_+ , \end{cases}$$

with

$$k_H = \pm(\omega - m\Omega_H) .$$

Wronskian Conservation

For Differential Equation

$$y'' + p(x)y' + q(x)y = 0$$

Define Wronskian

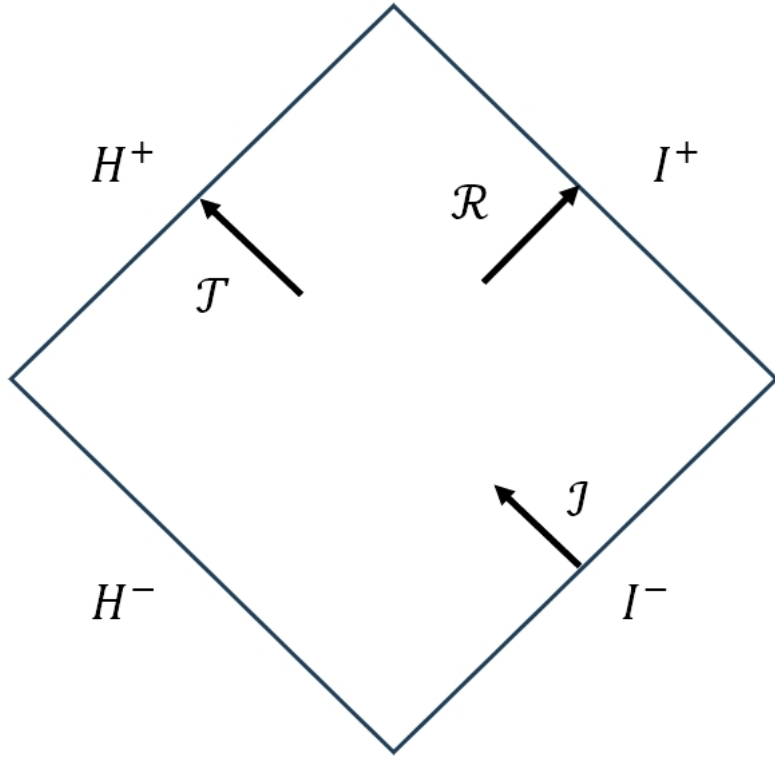
$$W = y_1 y_2' - y_2 y_1' \quad \text{so} \quad W' = -pW$$

For our EoM

$$\left[\frac{d^2}{dr_*^2} + V_{eff} \right] \chi(r_*) = 0$$

$$p = 0 \quad \longrightarrow \quad W' = 0 \quad \longrightarrow \quad \text{Wronskian Conserve}$$

Superradiant Scattering



Wronskian conservation:

$$W = \chi \frac{d\chi^*}{dr} - \chi^* \frac{d\chi}{dr} ,$$

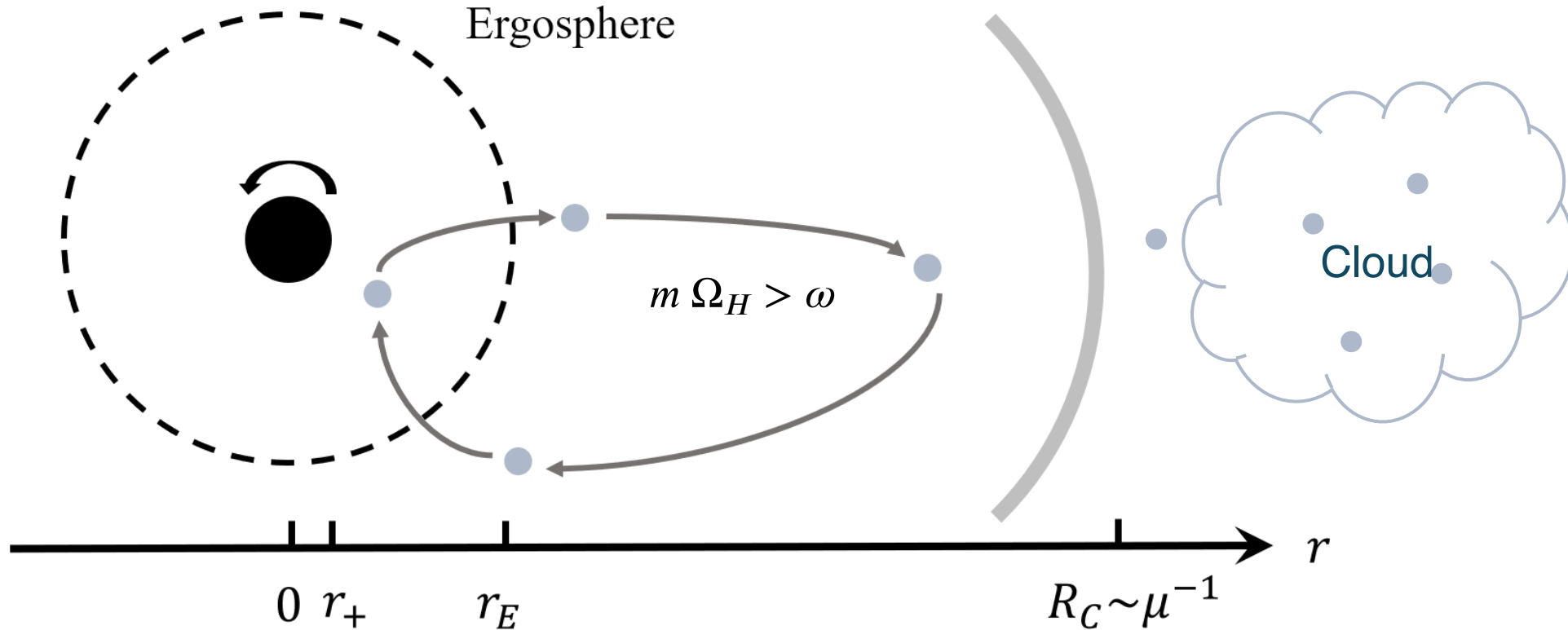


$$|\mathcal{R}|^2 + \frac{\omega - m\Omega_H}{\omega} |\mathcal{T}|^2 = 1$$

When $m\Omega_H > \omega$, **superradiant!**

Superradiant Instability

[Press & Teukolsky, 1972]
[Damour et al. 1976]

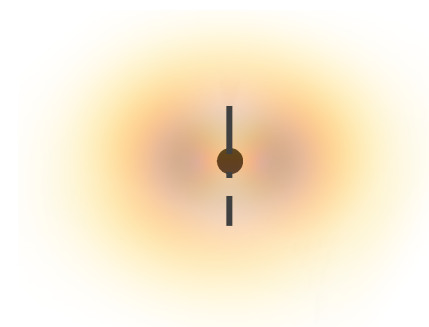
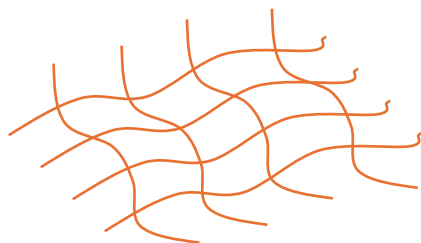
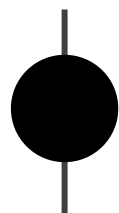


- Superradiant instability



Kerr BH grows an ultralight bosonic cloud

BH as a GA



Kerr BH of mass M

Bosonic field of mass μ

Gravitational Atom

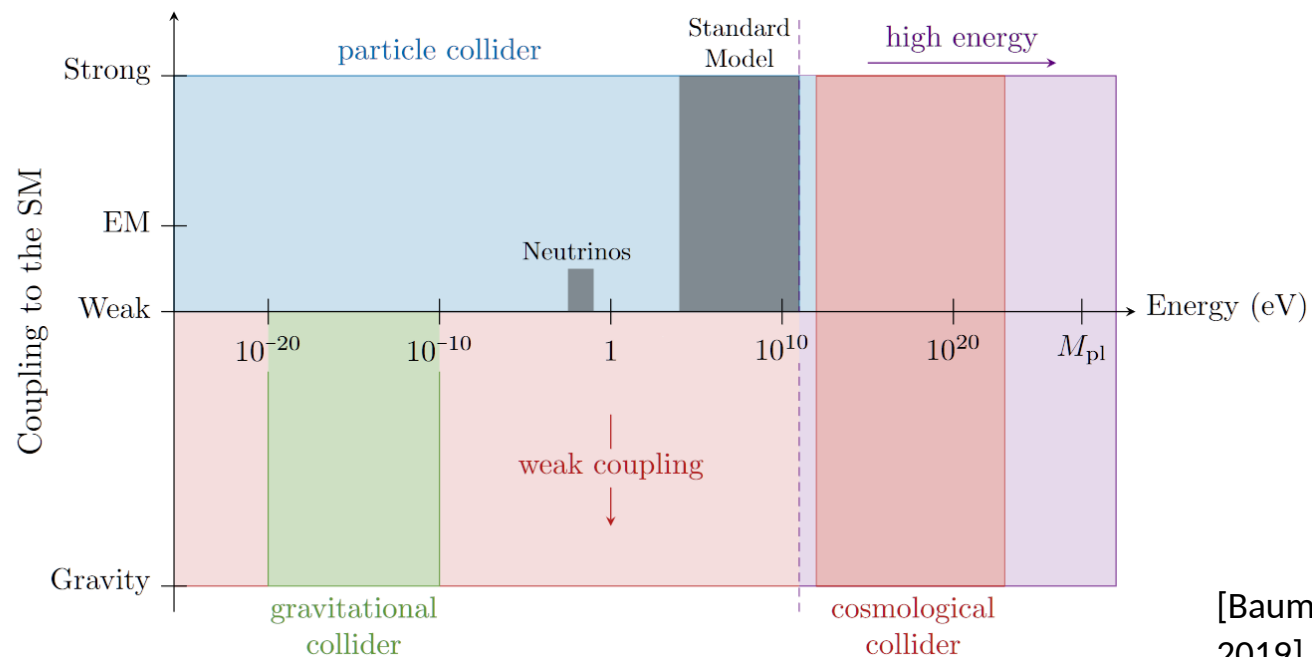
$$\mu^{-1} > GM$$

Compton
wavelength

Black hole
radius

BH

Ke



[Baumann et al, 2019]

$$\mu^{-1} > GM$$

$$M_B \sim [10M_{\odot}, 10^{10}M_{\odot}]$$

Compton
wavelength

Black hole
radius

BH as a GA

KG in Kerr:

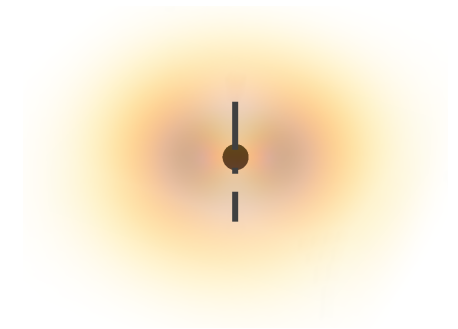
$$(g^{\alpha\beta} \nabla_\alpha \nabla_\beta - \mu^2) \Phi = 0$$

Gravitational fine structure constant

$$\alpha \equiv GM\mu \sim \frac{R_S}{\lambda_C} < 1$$

Factoring out the rest mass

$$\Phi \equiv \frac{1}{\sqrt{2\mu}} e^{-i\mu t} \psi + \text{c.c.}$$



Gravitational Atom

Hydrogen-like Schrodinger eq

$$i\partial_t \psi(t, \mathbf{r}) = H_0 \psi(t, \mathbf{r}) , \quad H_0 \equiv -\frac{1}{2\mu} \partial_{\mathbf{r}}^2 + V(r) \quad \text{with} \quad V(r) = -\frac{\alpha}{r} + \mathcal{O}(\alpha^2)$$

[Press & Teukolsky, 1972]

[Damour et al., 1976]

[Detweiler, 1980]

[Baumann et al, 2019, 2020]

GA in a nutshell

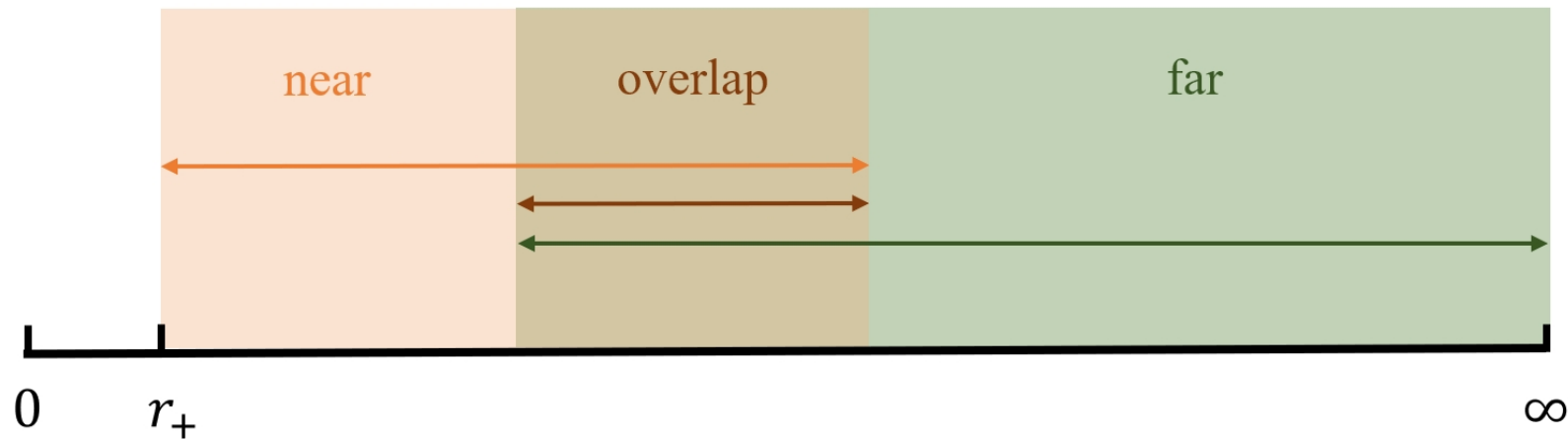
[Press & Teukolsky, 1972]
[Damour et al., 1976]
[Detweiler, 1980]
[Baumann et al, 2019, 2020]

- With **decaying B.C.** at infinity
- With **in-going B.C.** at the BH outer horizon

Solutions:

$|\psi_{nlm}\rangle$ with $\omega_{nlm} = E_{nlm} + i\Gamma_{nlm}$ ← By in-going B.C.

Determined by near far formalism :



GA in a nutshell

- With **decaying B.C.** at infinity
- With **in-going B.C.** at the BH outer horizon

Solutions:

$|\psi_{nlm}\rangle$ with $\omega_{nlm} = E_{nlm} + \boxed{i\Gamma_{nlm}}$ ← By in-going B.C.

$$E_{nlm} = \mu \left(1 - \frac{\alpha^2}{2n^2} + \alpha^4 A(n, l) + \alpha^5 \tilde{a} m B(n, l) + \dots \right)$$

Rest mass

Bohr

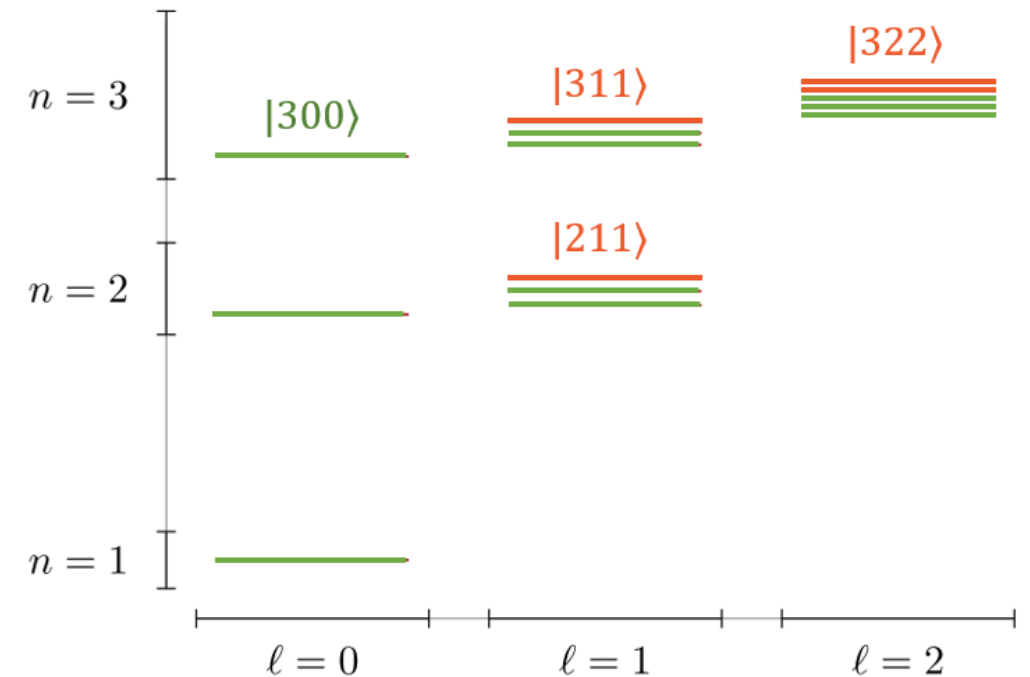
Fine

Hyperfine

$$\Gamma_{nlm} \propto (m\Omega_H - \mu)\alpha^{4l+5} \begin{cases} > 0 & \text{Superradiance} \\ < 0 & \text{Absorption} \end{cases}$$

$$\psi_{nlm} \sim e^{-i\omega_{nlm}t} \sim e^{\Gamma_{nlm}t}$$

[Press & Teukolsky, 1972]
 [Damour et al., 1976]
 [Detweiler, 1980]
 [Baumann et al, 2019, 2020]

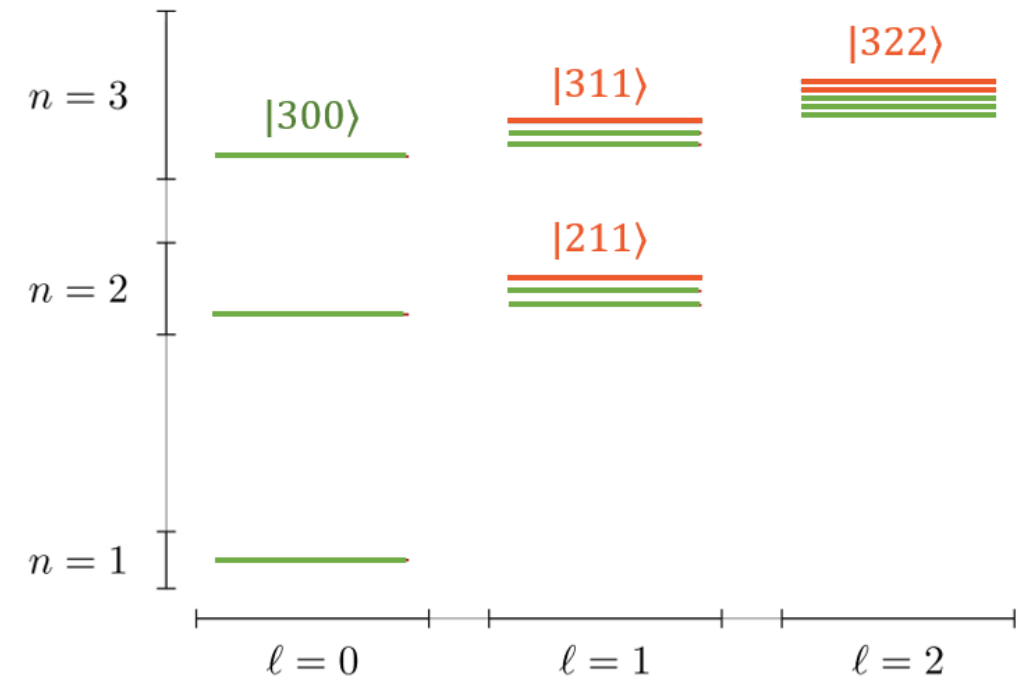


GA in a nutshell

[Press & Teukolsky, 1972]
 [Damour et al., 1976]
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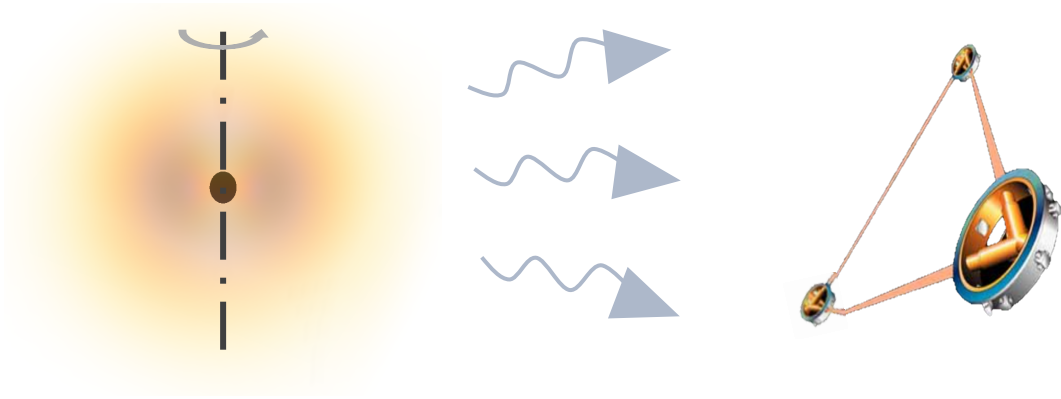
$$\Gamma_{nlm} \propto (m\Omega_H - \mu)\alpha^{4l+5} \quad \text{with} \quad \alpha \equiv GM\mu \sim \frac{R_S}{\lambda_C} < 1$$

- From $m\Omega_H - \mu$
 - To make sure it is bigger than zero, the spin should high enough
 - Cloud saturated when spin is small enough.
 - $m \leq 0$ mode always absorptive.
- From α^{4l+5} :
 - The lower l modes grow much quicker.
 - $|211\rangle$ mode will dominant first if spin is enough.

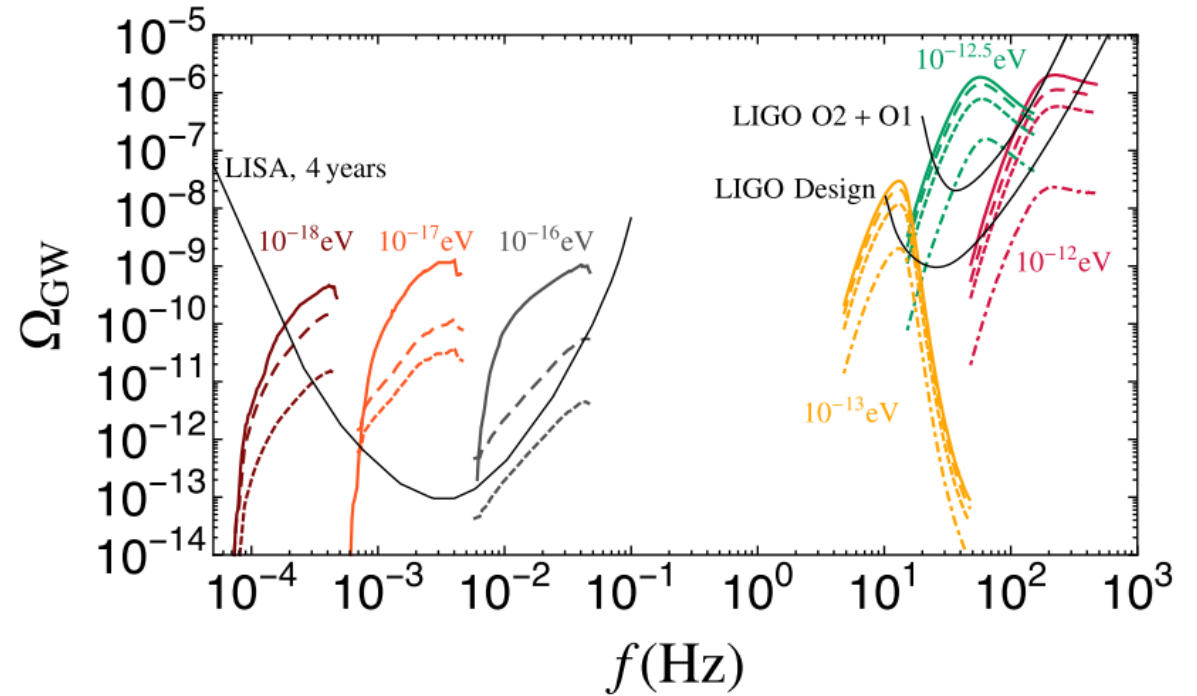


GA phenomenology in isolation

- Near-monochromatic GW



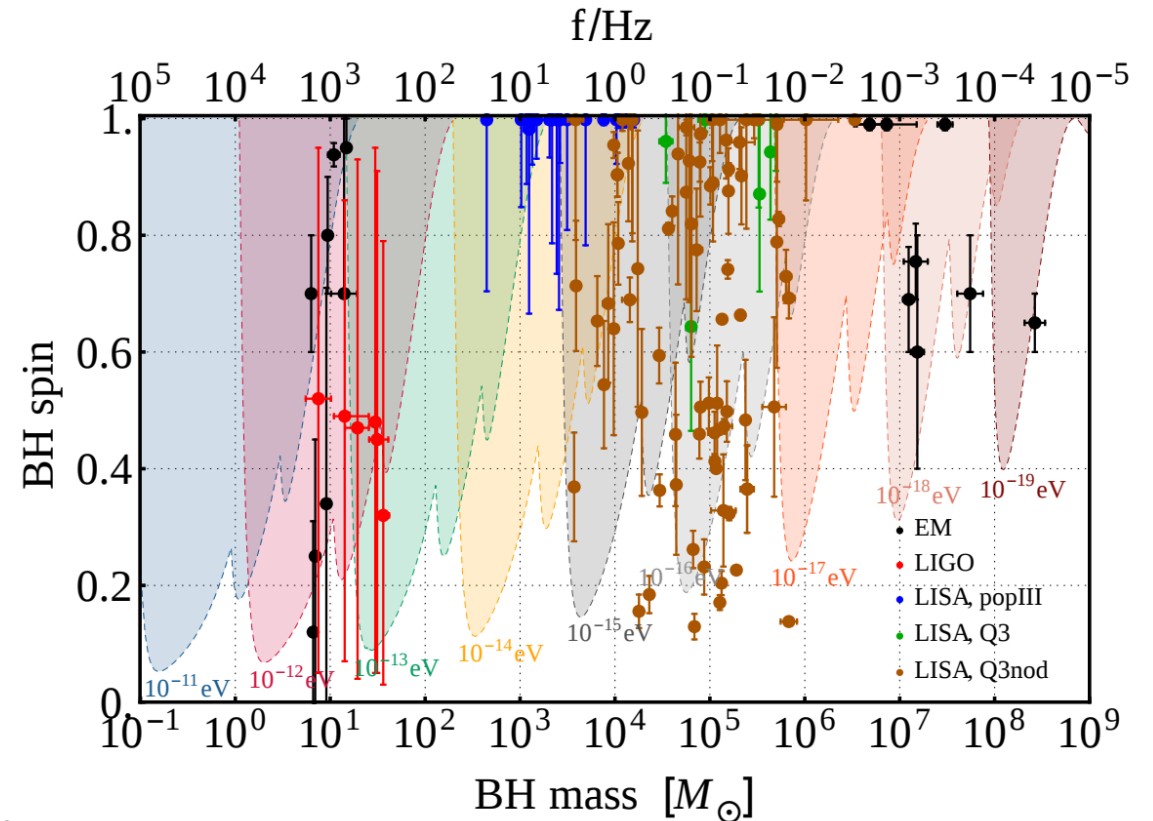
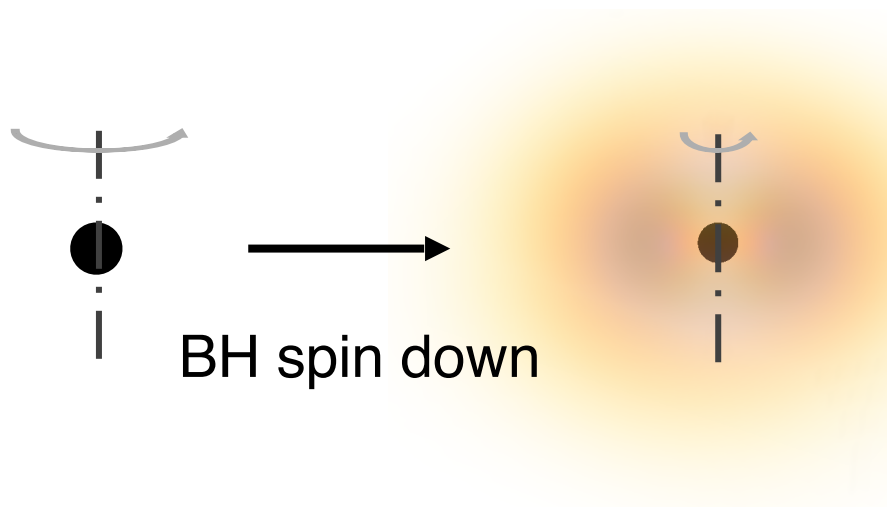
$$f_{\text{GW}} \sim \omega_R / \pi \sim 5 \text{ kHz} \left(\frac{\mu \hbar}{10^{-11} \text{ eV}} \right)$$



[Brito et al., 2017]

GA phenomenology in isolation

- Spin cutoff by superradiance



$$m\Omega_H \downarrow > \omega \sim \mu \quad \rightarrow \quad \frac{a}{M} = \frac{4m(M\omega)}{m^2 + 4(M\omega)^2} = \frac{4\alpha}{m} + \mathcal{O}(\alpha^3)$$

Spin at saturation

[Brito et al., 2017]

GA phenomenology in binaries

- Atomic resonant transitions $i\partial_t\psi(t, r) = H\psi(t, r)$, with $H = H_0 + V_*(t)$

$$V_* = -\alpha q \sum_{l_* \geq 2} \sum_{|m_*| \leq l_*} \mathcal{E}_{l_* m_*}(\iota_*, \varphi_*) Y_{l_* m_*}(\theta, \phi) \times \left(\frac{r^{l_*}}{R_*^{l_*+1}} \Theta(R_* - r) + \frac{R_*^{l_*}}{r^{l_*+1}} \Theta(r - R_*) \right)$$

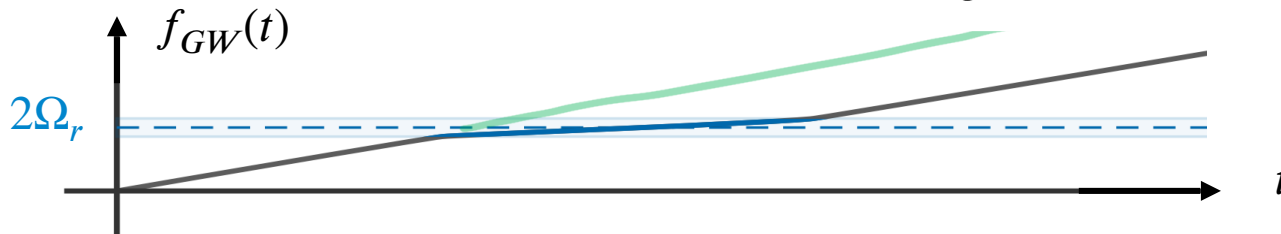
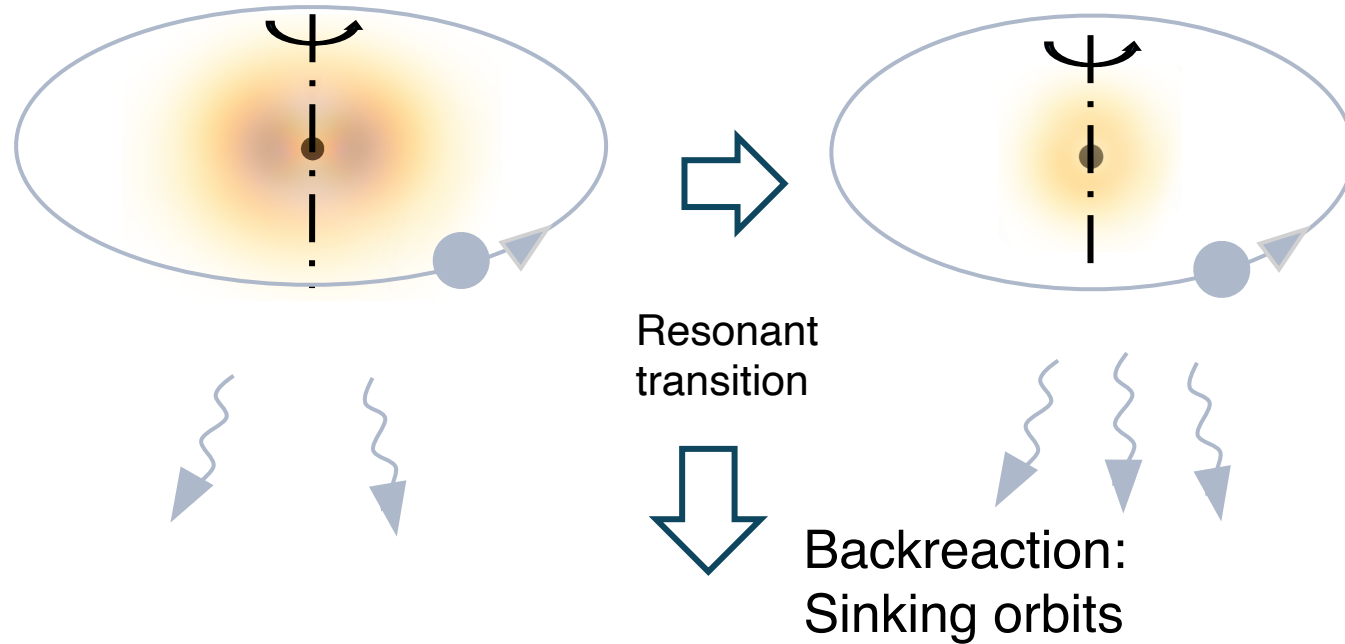
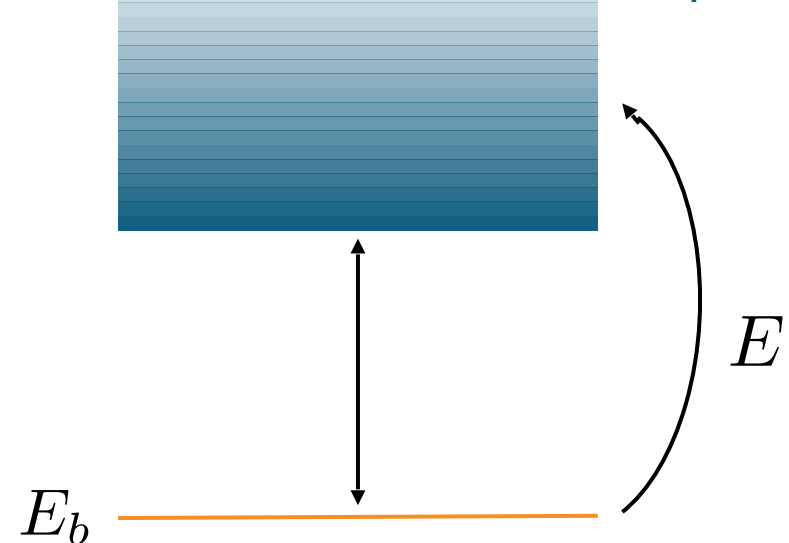


Figure adapted from [Baumann et al, 2020]

Unbound states:
Continuous spectrum



GA phenomenology in binaries

- Atomic resonant transitions $i\partial_t\psi(t, r) = H\psi(t, r)$, with $H = H_0 + V_*(t)$

Ionization or Dynamical Friction

$$P_{\text{DF}} = \frac{4\pi q^2 M_{\text{BH}}^2}{v} \rho_{nlm}(R_*, \varphi_*) C_{\Lambda}$$



Backreaction:
Sinking orbits

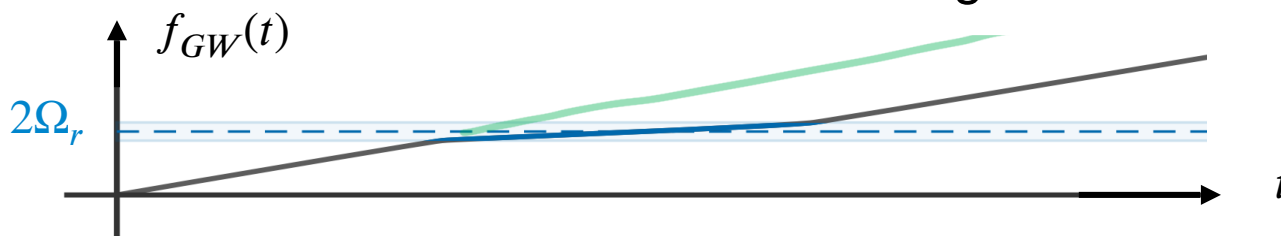
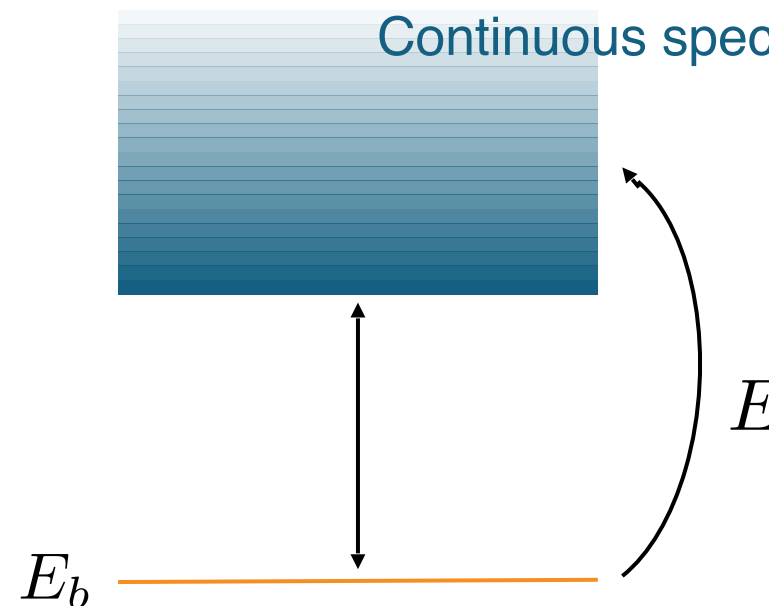


Figure adapted from [Baumann et al, 2020]

$$V_* = -\alpha q \sum_{l_* \geq 2} \sum_{|m_*| \leq l_*} \mathcal{E}_{l_* m_*}(\iota_*, \varphi_*) Y_{l_* m_*}(\theta, \phi) \\ \times \left(\frac{r^{l_*}}{R_*^{l_*+1}} \Theta(R_* - r) + \frac{R_*^{l_*}}{r^{l_*+1}} \Theta(r - R_*) \right)$$

Unbound states:
Continuous spectrum





Combining Observation In Reality.

OJ 287

$$M_{\text{BH}} = 1.8 \times 10^{10} M_{\odot}$$

$$m_{\text{BH}} = 1.5 \times 10^8 M_{\odot}$$

$$T = 12.067 \text{ yr}$$

$$\dot{T} = -0.00099$$

OJ 287

[Valtonen and Lehto, 1997]
[Valtonen and Pihajoki, 2013]
[M. J. Valtonen et al, 2016]
[L. Dey et al., 2018]

Parameters	Value
Primary SMBH mass ($10^{10} M_{\odot}$)	1.8348 ± 0.0008
Secondary SMBH mass ($10^8 M_{\odot}$)	1.5013 ± 0.0025
Eccentricity e	0.657 ± 0.001
Precession angle per period $\Delta\vartheta$	$38.62^{\circ} \pm 0.01^{\circ}$
Orbital period T (yr)	12.067 ± 0.007
Orbital period decay rate $\dot{T}$	-0.00099 ± 0.00006
Spin of the primary SMBH χ_{P_2}	0.381 ± 0.004
Spin of the primary SMBH χ_{P_1}	0.313 ± 0.01
Semi-major axis a (pc)	0.056
Inclination angle of orbit plane	90°

OJ 287

$$P_{\text{orb}} = -\frac{GMm}{3a} \frac{\dot{T}}{T} = (3.66 \pm 0.24) \times 10^{41} \text{ W}$$

$$P_{\text{GW}} = \frac{32}{5} \frac{G^4 M^2 m^2 (M + m)}{c^5 a^5} f(e) = (2.62 \pm 0.02) \times 10^{41} \text{ W}$$

OJ 287

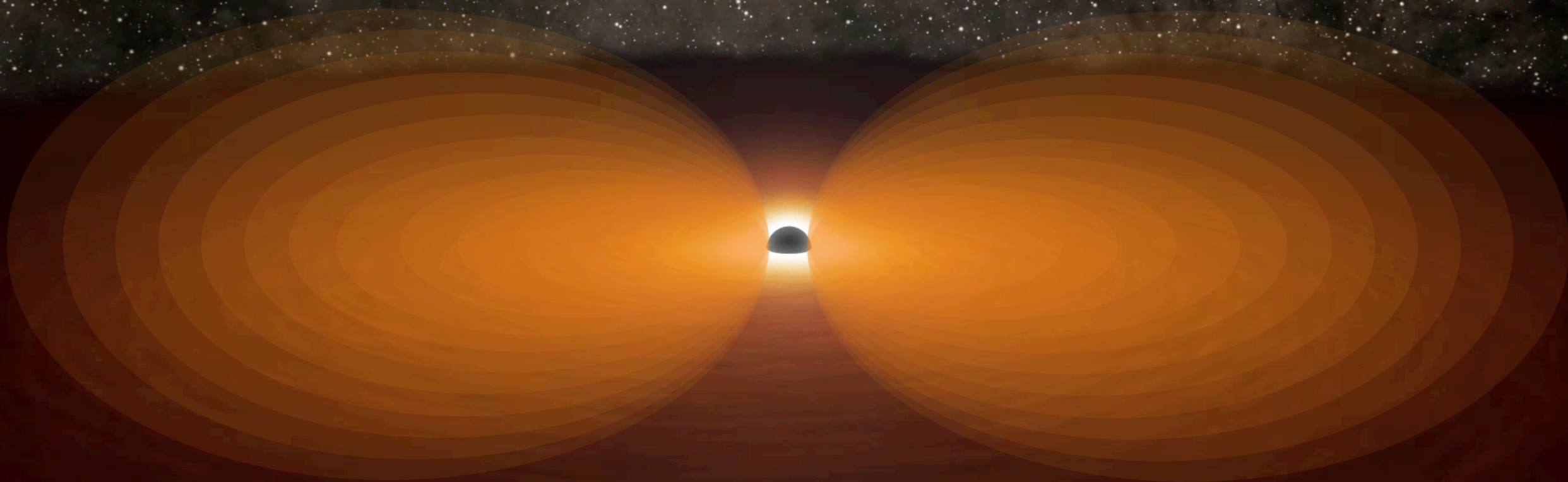
$$P_{\text{orb}} = -\frac{GMm}{3a} \frac{\dot{T}}{T} = (3.66 \pm 0.24) \times 10^{41} \text{ W}$$

4.3σ

$$P_{\text{GW}} = \frac{32}{5} \frac{G^4 M^2 m^2 (M + m)}{c^5 a^5} f(e) = (2.62 \pm 0.02) \times 10^{41} \text{ W}$$

OJ 287

$$P_{\text{orb}} = P_{\text{GW}} + P_{\text{DF}}$$





Is the cloud stably exist ?

GA phenomenology in binaries

- Atomic resonant transitions

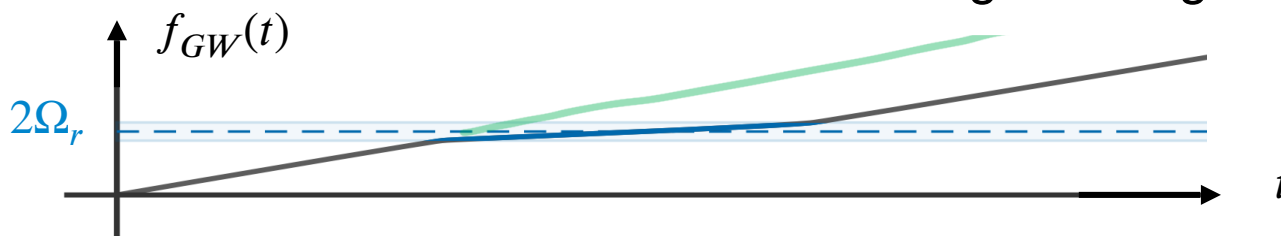
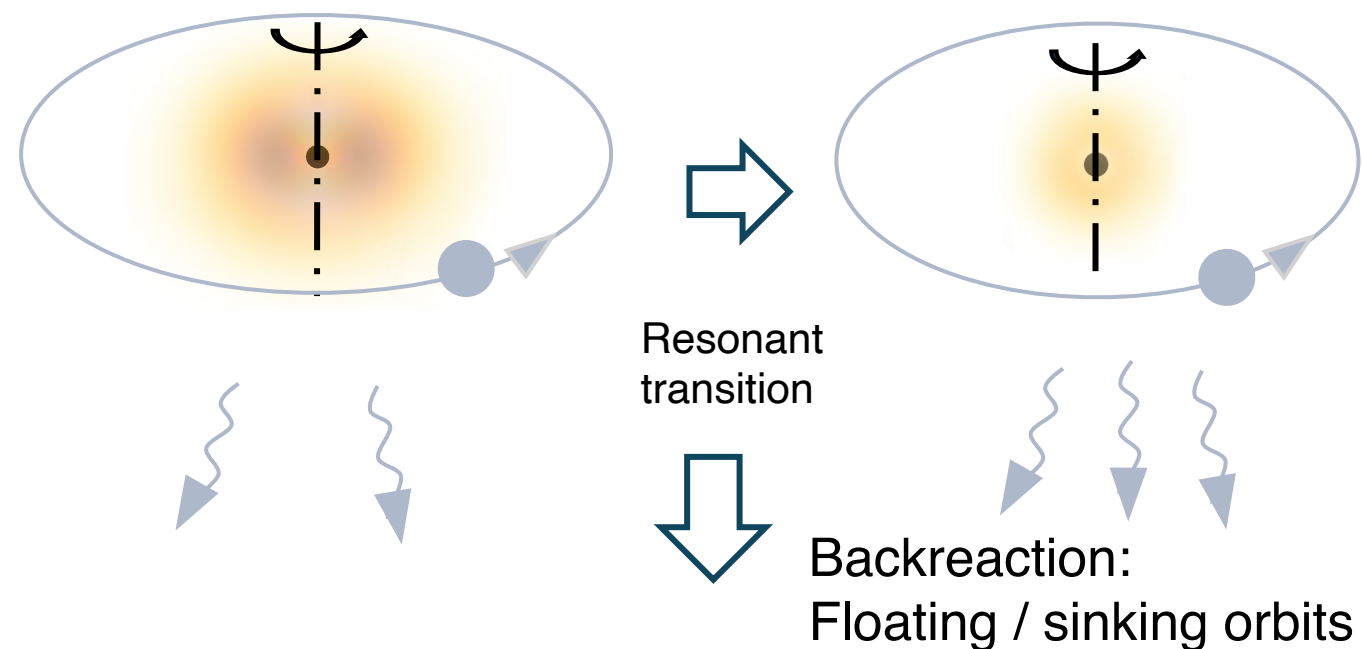
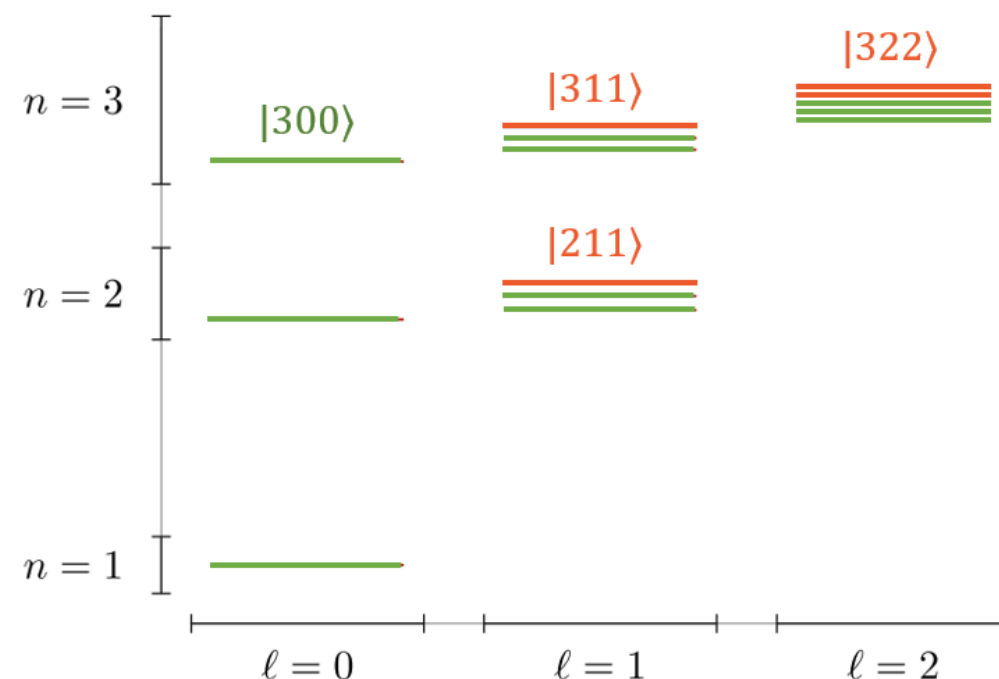


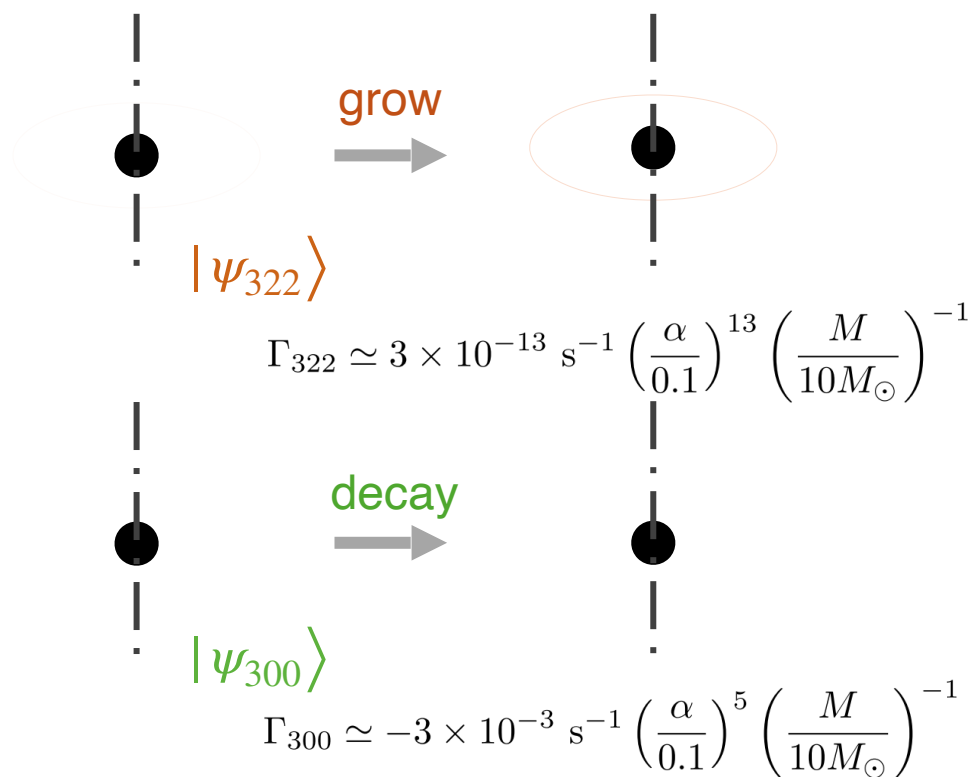
Figure adapted from [Baumann et al, 2020]



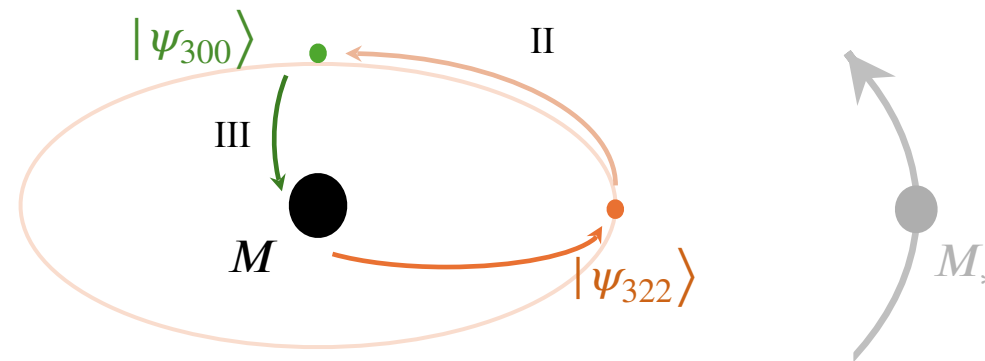
GA phenomenology in binaries

[Tong, Wang & Zhu, 2022]

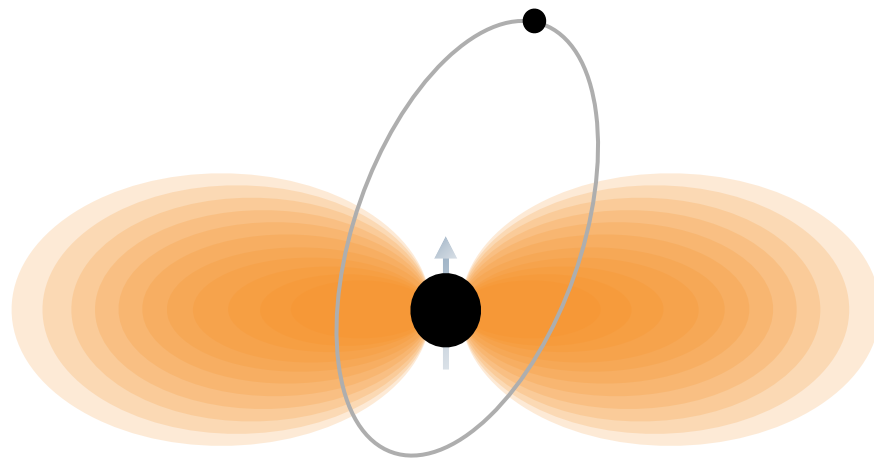
GA in **isolation**



GA in a **binary**



OJ 287

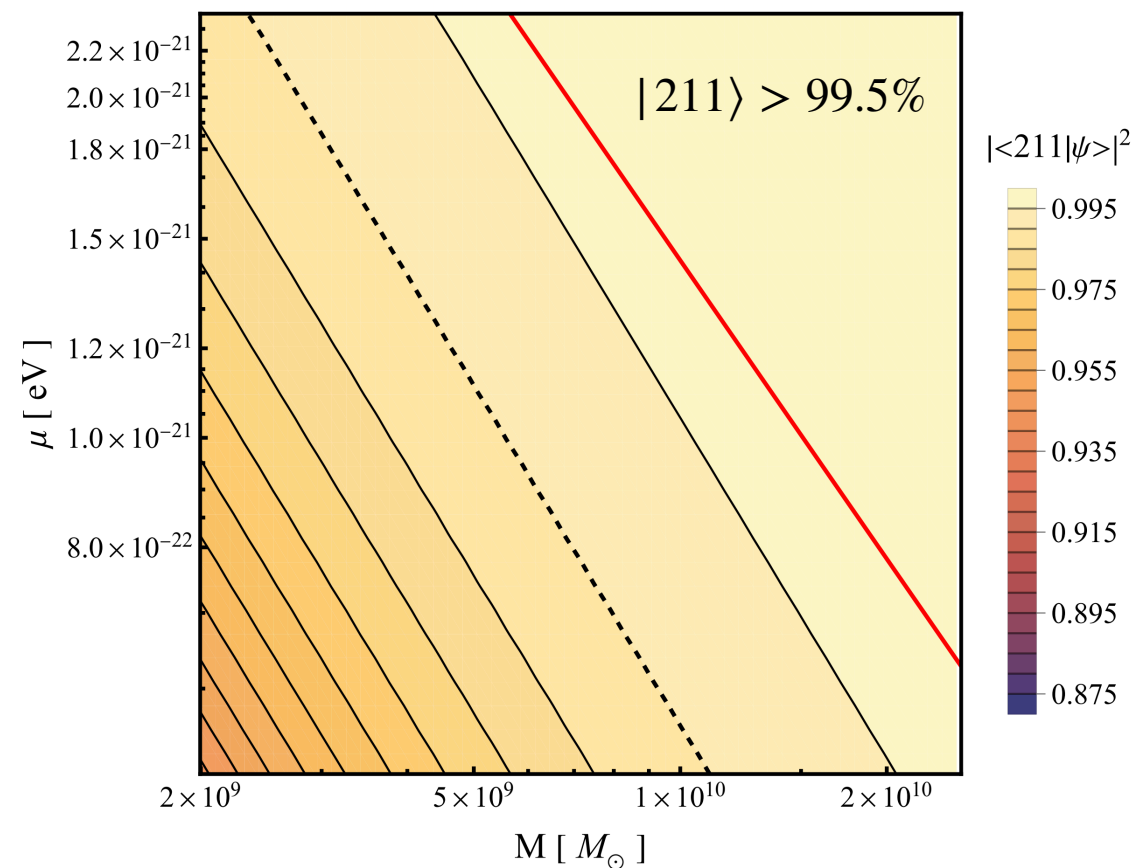


- The state $|211\rangle$ has saturated.
- The cloud can be survived from bound-to-bound with survival rate bigger than 97.5%.

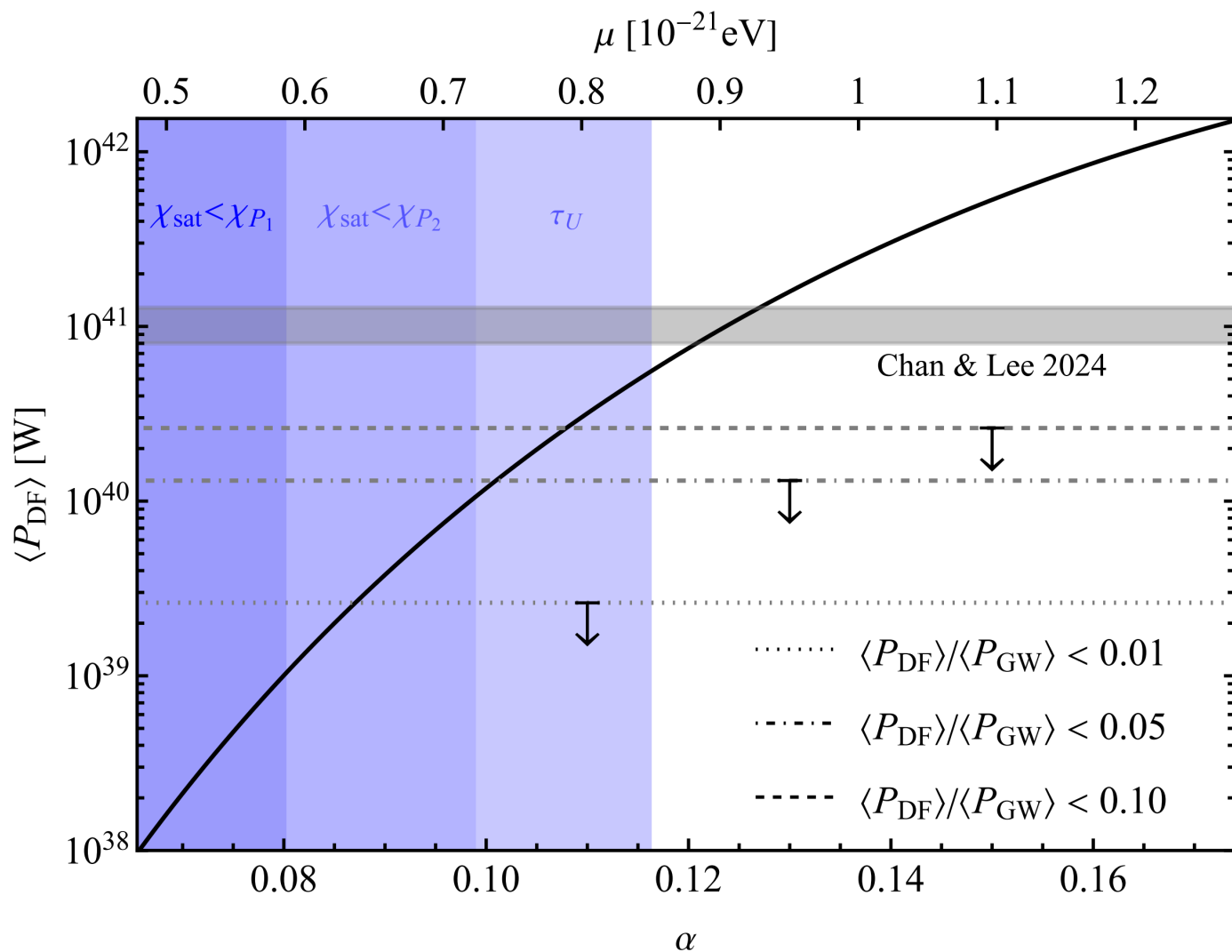
$$P_{\text{DF}} = \frac{4\pi M_*^2}{v} \rho_{nlm} C_\Lambda$$

ρ_{211} by solving KG equation

$$\langle P_{\text{DF}} \rangle = \frac{1}{T} \int_0^{2\pi} P_{\text{DF}}(R_*(\varphi_*), \theta_*(\varphi_*, \iota)) \frac{dt}{d\varphi_*} d\varphi_*$$



OJ 287 Constraint on Ultralight Boson



Dissipative Energy from Boson Cloud

$$P_{\text{orb}} = P_{\text{GW}} + P_{\text{DF}}$$

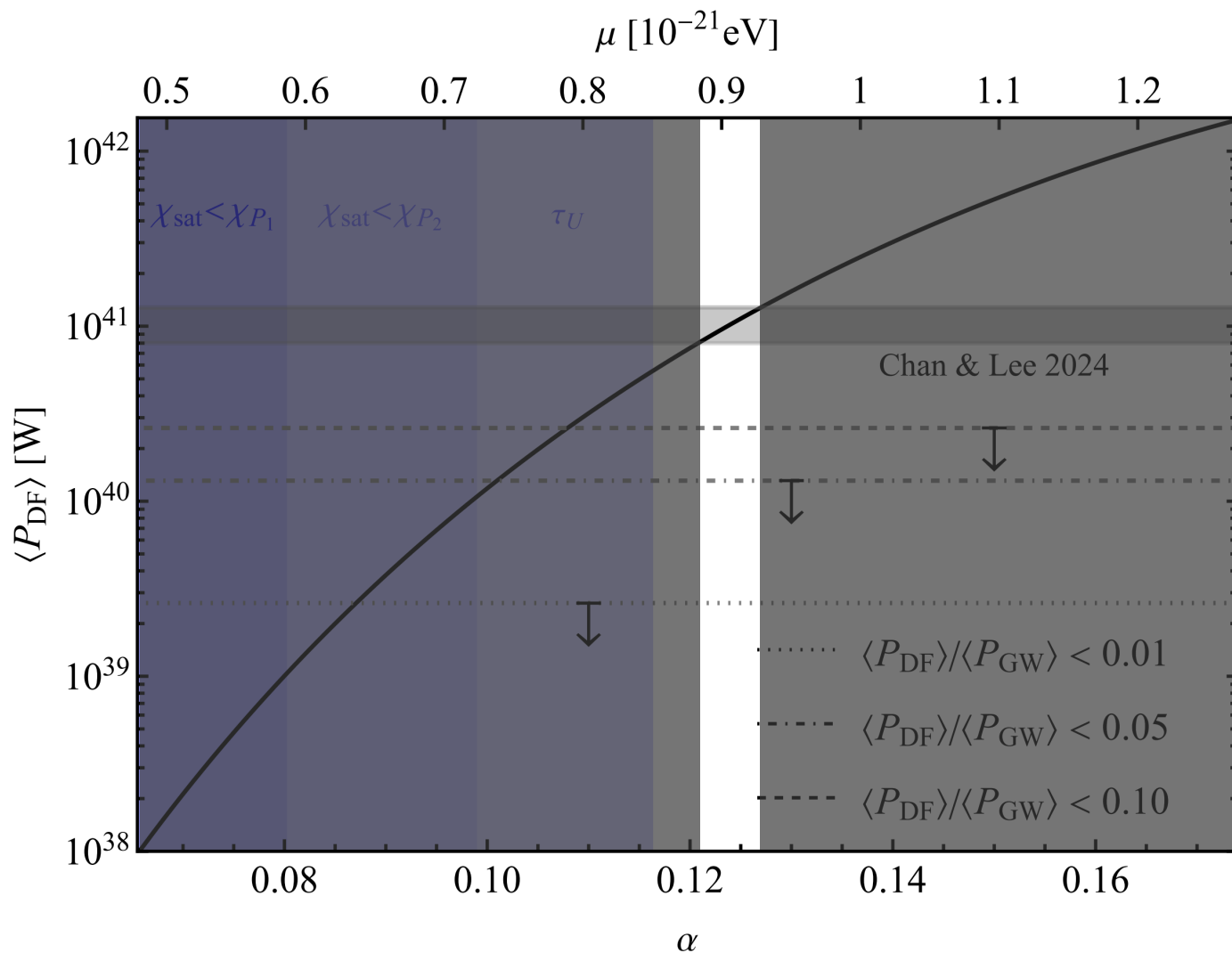
$$\mu = (8.8 - 9.3) \times 10^{-22} \text{ eV}$$

A Null Detection Result

$$\frac{P_{\text{DF}}}{P_{\text{GW}}} < 1\%$$

$$\mu \notin (8.5 - 22) \times 10^{-22} \text{ eV}$$

OJ 287 Constraint on Ultralight Boson



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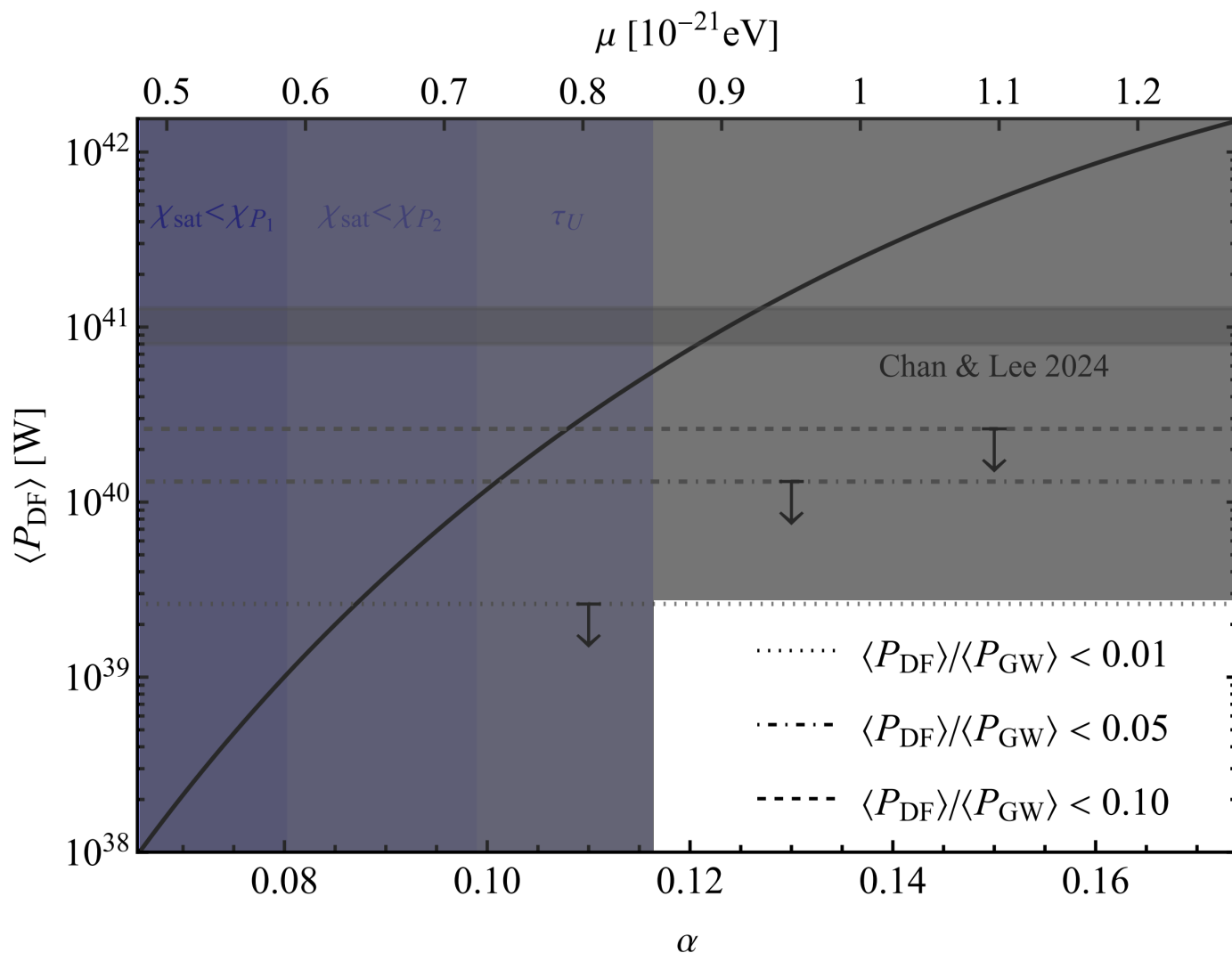
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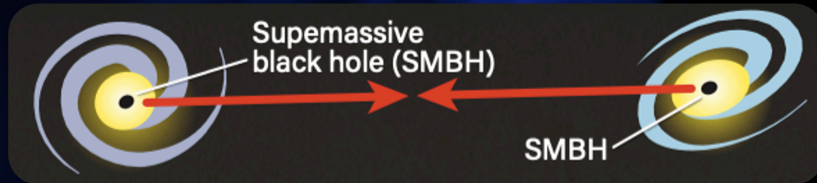
$$\frac{P_{\text{DF}}}{P_{\text{GW}}} < 1\%$$

$$\mu \notin (8.5 - 22) \times 10^{-22} \text{ eV}$$



What can we do other than OJ287 ?

Final Parsec Problem



1. Two galaxies start merging



2. SMBH binary shrinks via flung away star and gas.

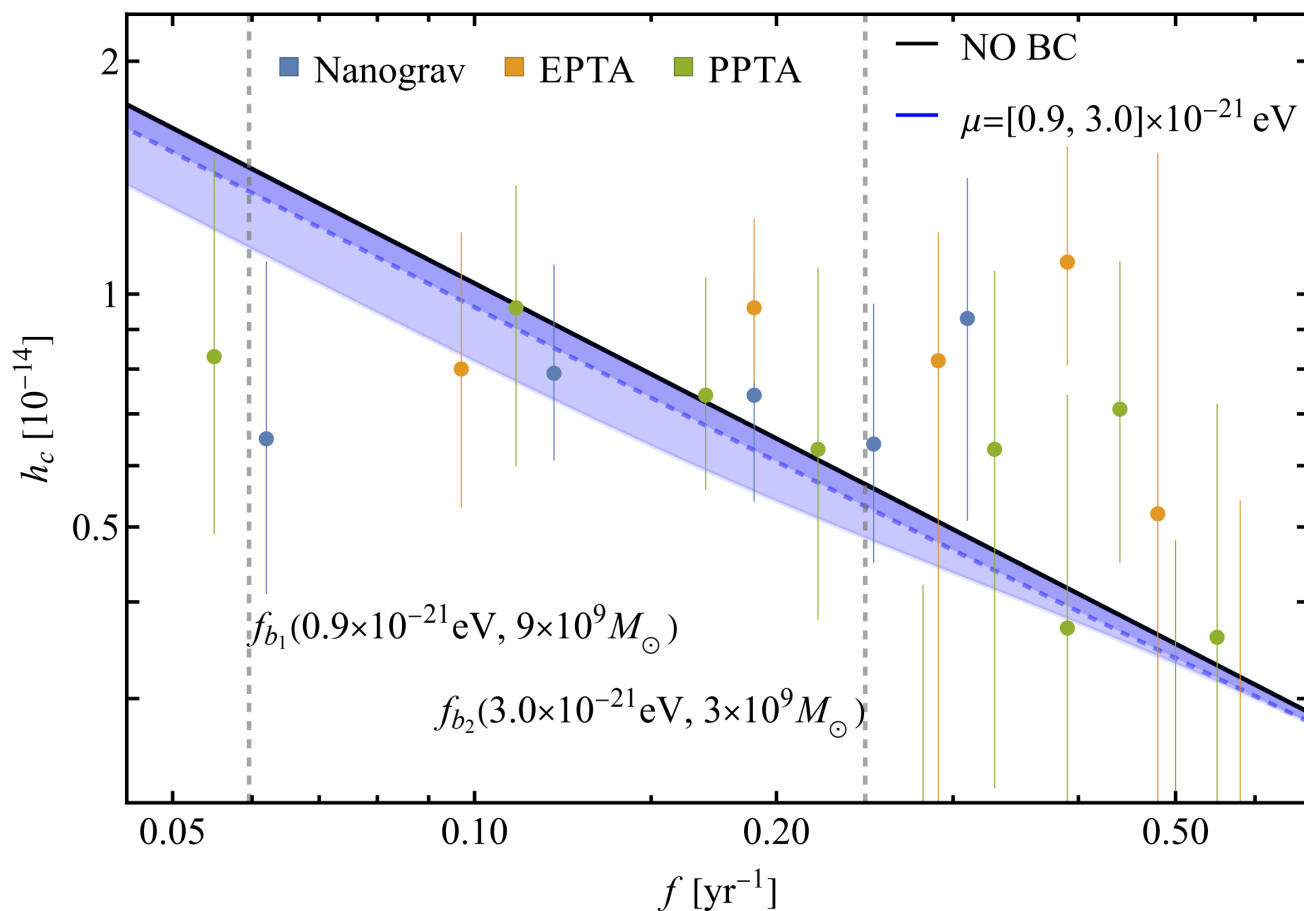


3. No surrounding stars cause SMBH binary stalling at final parsec more than the age of universe.



4. SMBH binary is driven by GW at last 0.01 parsec until merger.

Stochastic Gravitational Wave Background

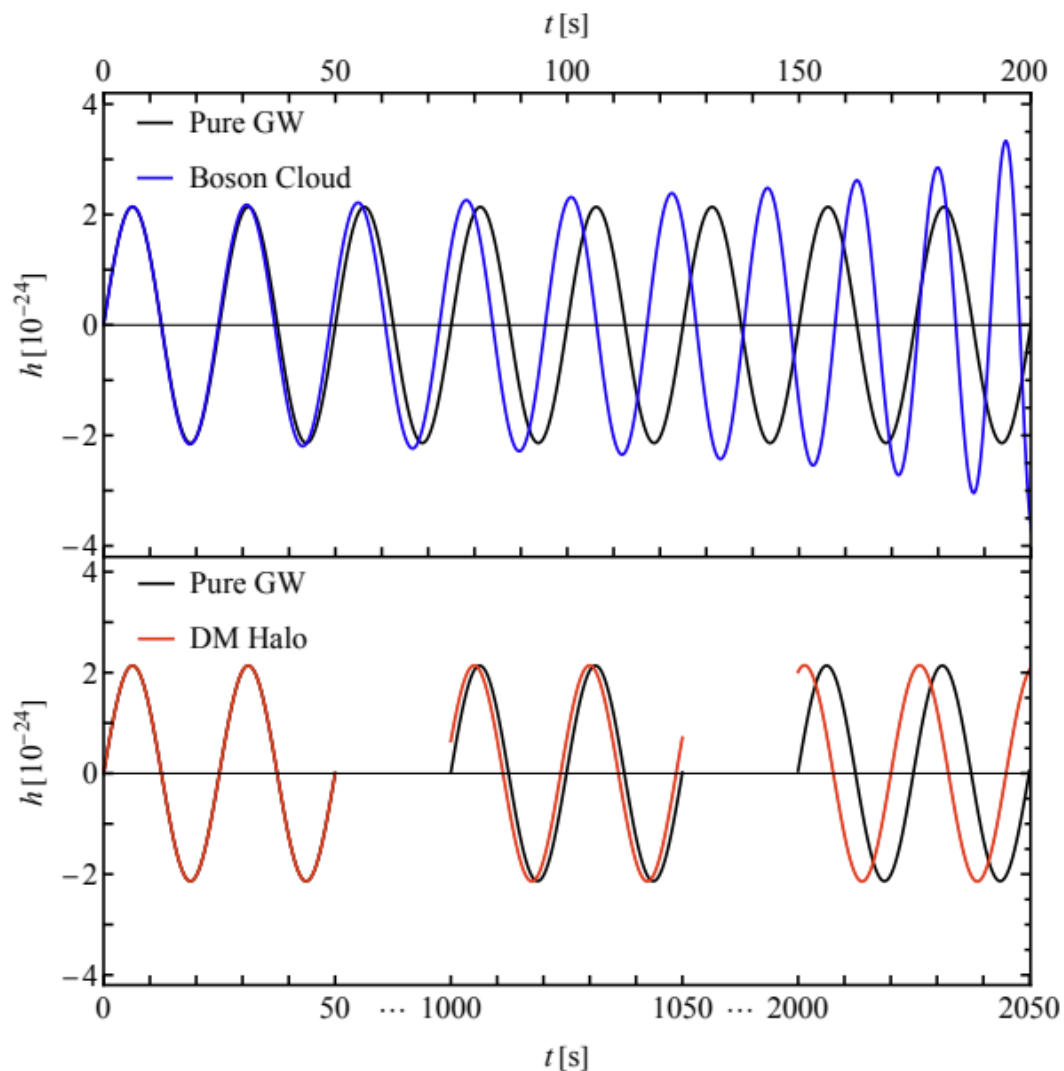


- After crossing final parsec, GW emission dominates and sources nanohertz.

$$\left\{ \begin{aligned} \frac{dE_{\text{GW}}}{df_s} &= \frac{\pi^{2/3}}{3} \frac{M_{\text{BH}}^{5/3}}{f_s^{1/3}} \frac{q}{(1+q)^{1/3}} \frac{\langle P_{\text{GW}} \rangle}{\langle P_{\text{GW}} \rangle + \langle P_{\text{DF}} \rangle} \\ h_c^2(f) &= \frac{4}{\pi f} \int dz dq dM_{\text{BH}} \frac{d^3 n}{dz dq dM_{\text{BH}}} \frac{dE_{\text{GW}}}{df_s} \end{aligned} \right.$$

$$f_b \simeq 3.2 \text{ nHz} \left(\frac{\mu}{10^{-21} \text{ eV}} \right)^3 \left(\frac{M_{\text{BH}}}{10^{10} M_\odot} \right)^2$$

Gravitational Wave Waveform



$$P_{\text{DF}} = \frac{4\pi q^2 M_{\text{BH}}^2}{v} \rho_{nlm}(R_*, \varphi_*) C_{\Lambda}$$

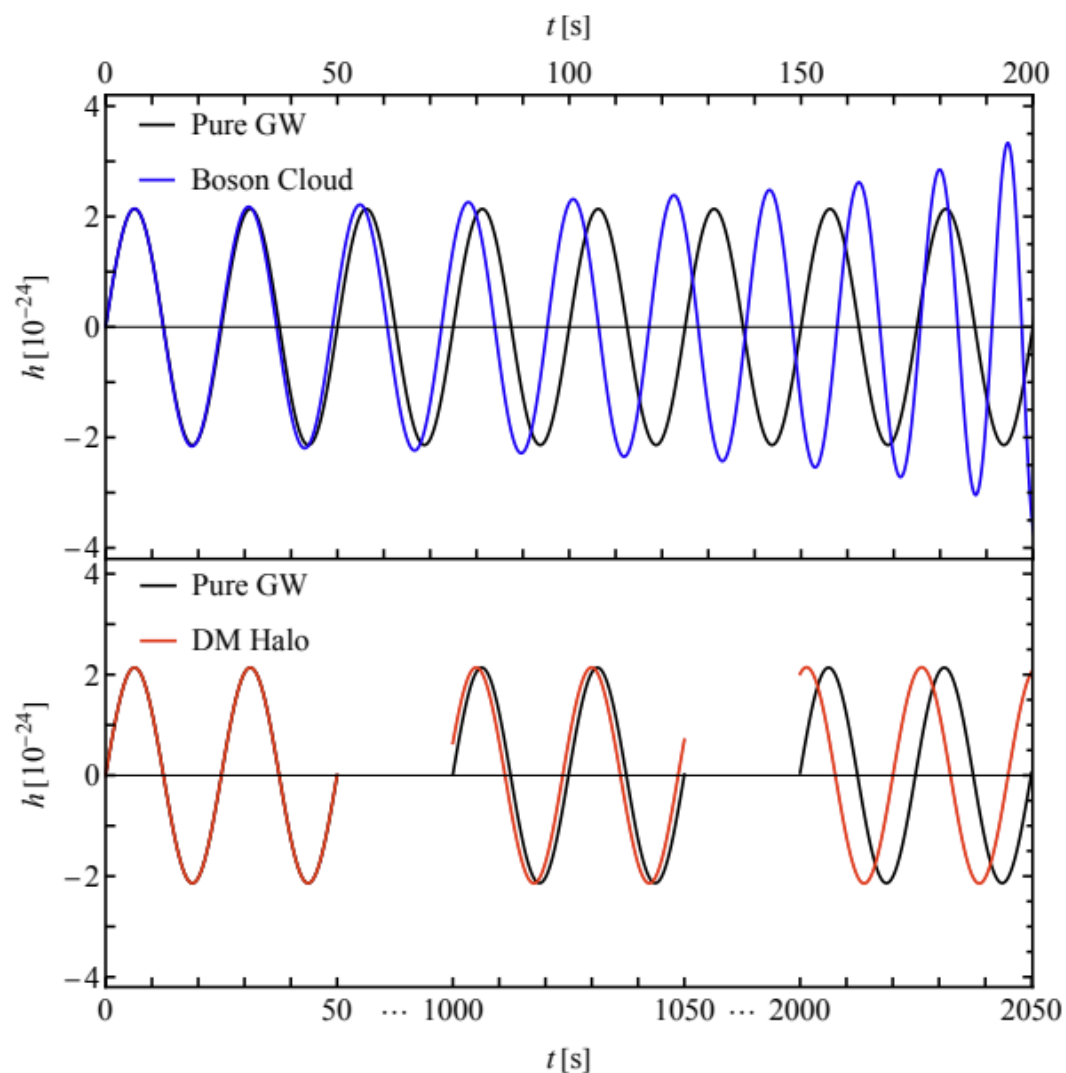
Dark Environment

GW dephasing



Can we construct a quantity using purely observable to characterize the BH environment?

Gravitational Wave Waveform

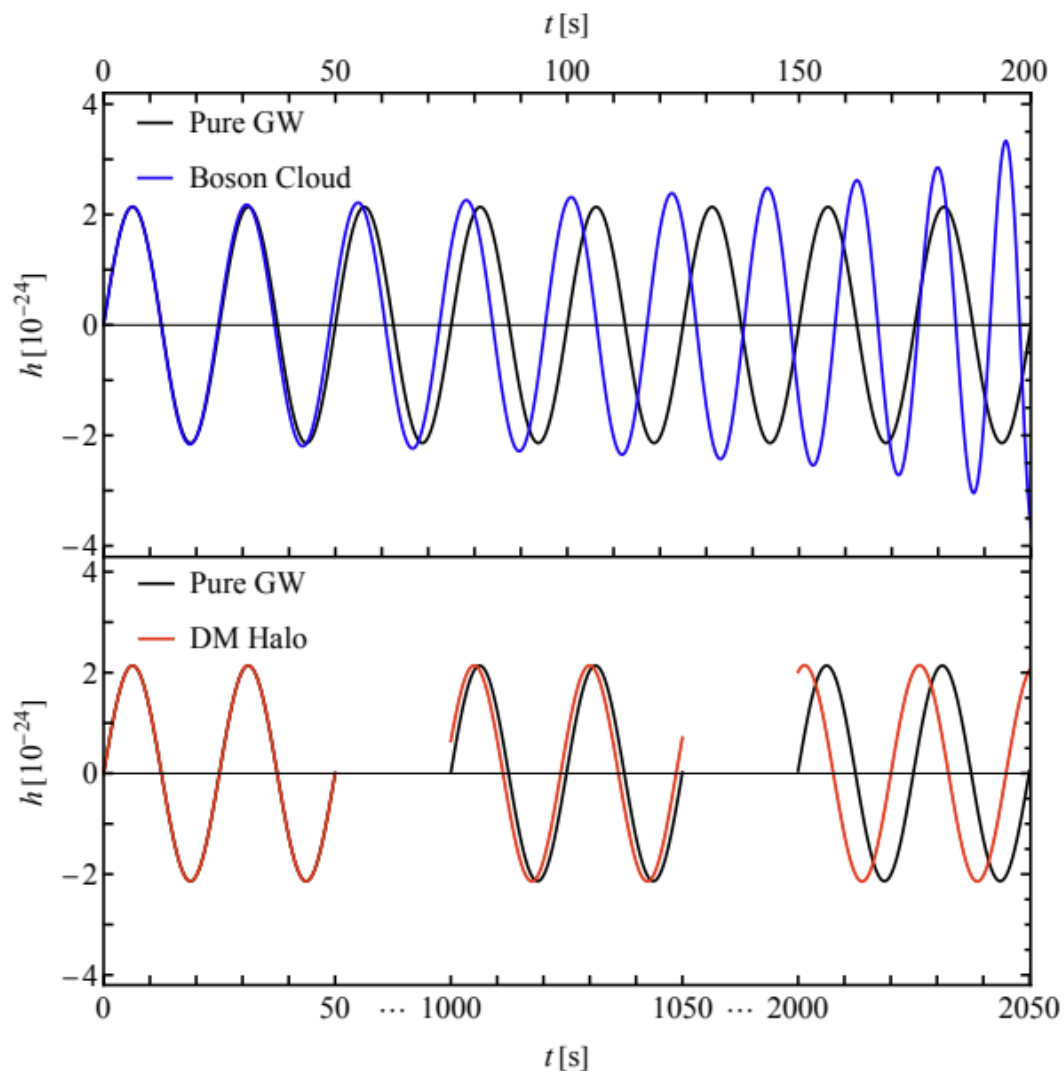


By GW Emission

$$\frac{df}{dt} = \frac{96 [G\mathcal{M}_c(1+z)]^{5/3} \pi^{8/3} f^{11/3}}{5 c^5} + \frac{3f^{1/3}}{(\pi G)^{2/3} [\mathcal{M}_c(1+z)]^{5/3}} |P_{\text{DF}}|$$

By DF extra power loss

Gravitational Wave Waveform



If no more than
GW

$$g(t) \equiv \frac{1}{hf^3} \frac{df}{dt} = \frac{24}{5} \frac{d_L(z)\pi^2}{c} + \frac{12G}{c^4 d_L(z)} \frac{|P_{\text{DF}}|}{h^2 f^2}$$

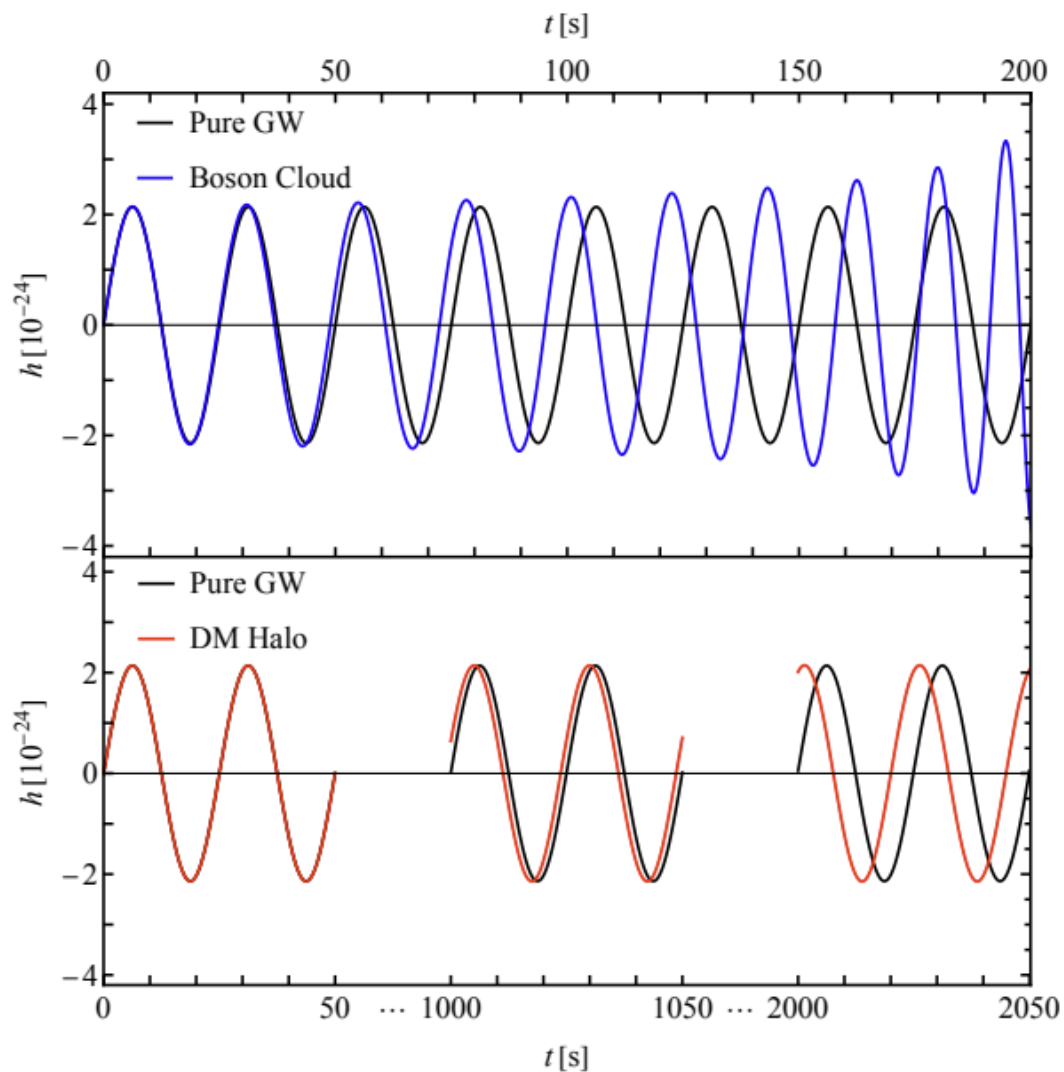


$$D \equiv -h^2 f^3 \frac{dg/dt}{df/dt}$$

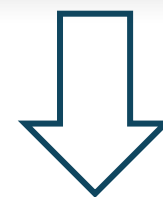
$$= \frac{12G}{c^4 d_L(z)} |P_{\text{DF}}| \left(\frac{11}{3} - \frac{d \ln \rho}{d \ln f} - \frac{d \ln C_\Lambda}{d \ln f} \right)$$

g becomes constant and $D = 0$

Gravitational Wave Waveform

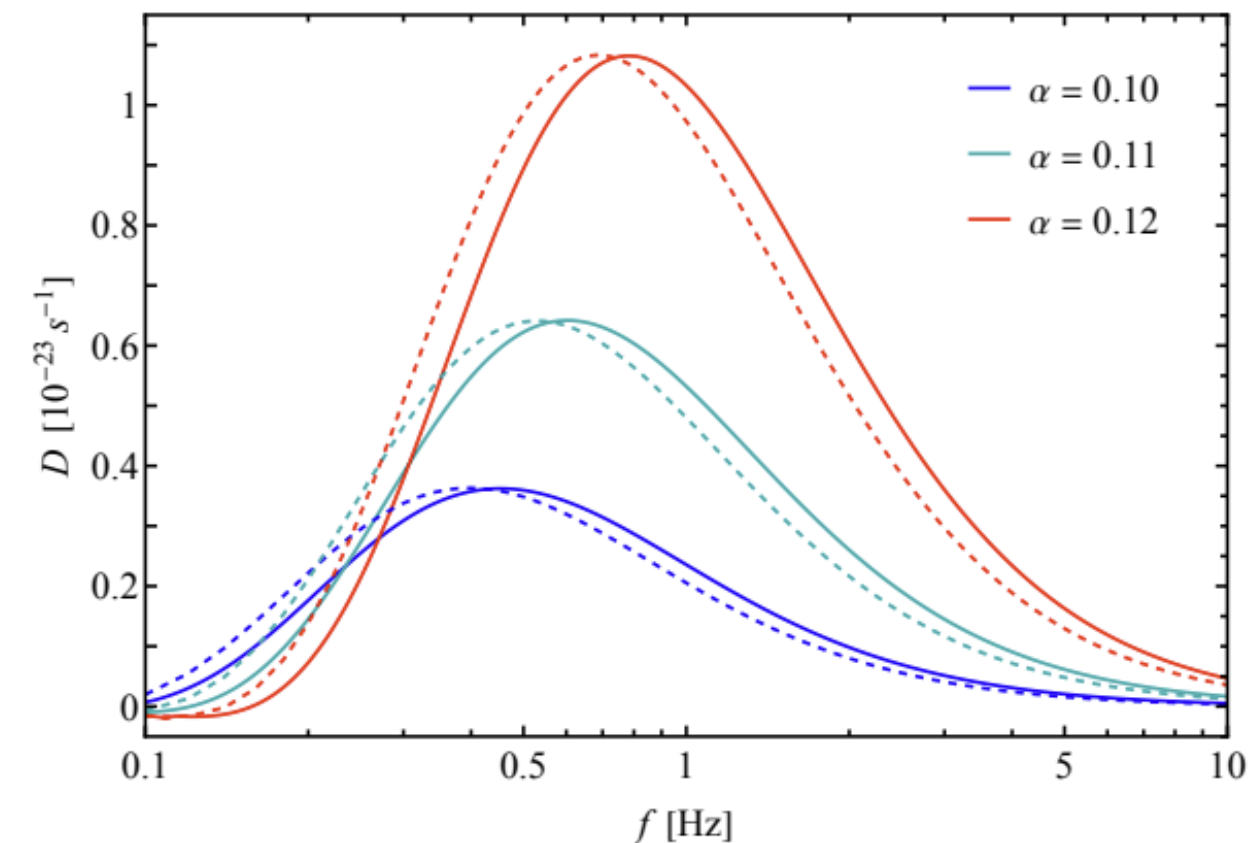


$$P_{\text{DF}} = -4\pi \frac{G^2 M_*^2}{v} \rho_D C_\Lambda$$



$$\begin{aligned}
 D &\equiv -h^2 f^3 \frac{dg/dt}{df/dt} \\
 &= \frac{12G}{c^4 d_L(z)} |P_{\text{DF}}| \left(\frac{11}{3} - \frac{d \ln \rho}{d \ln f} - \frac{d \ln C_\Lambda}{d \ln f} \right)
 \end{aligned}$$

Gravitational Wave Waveform



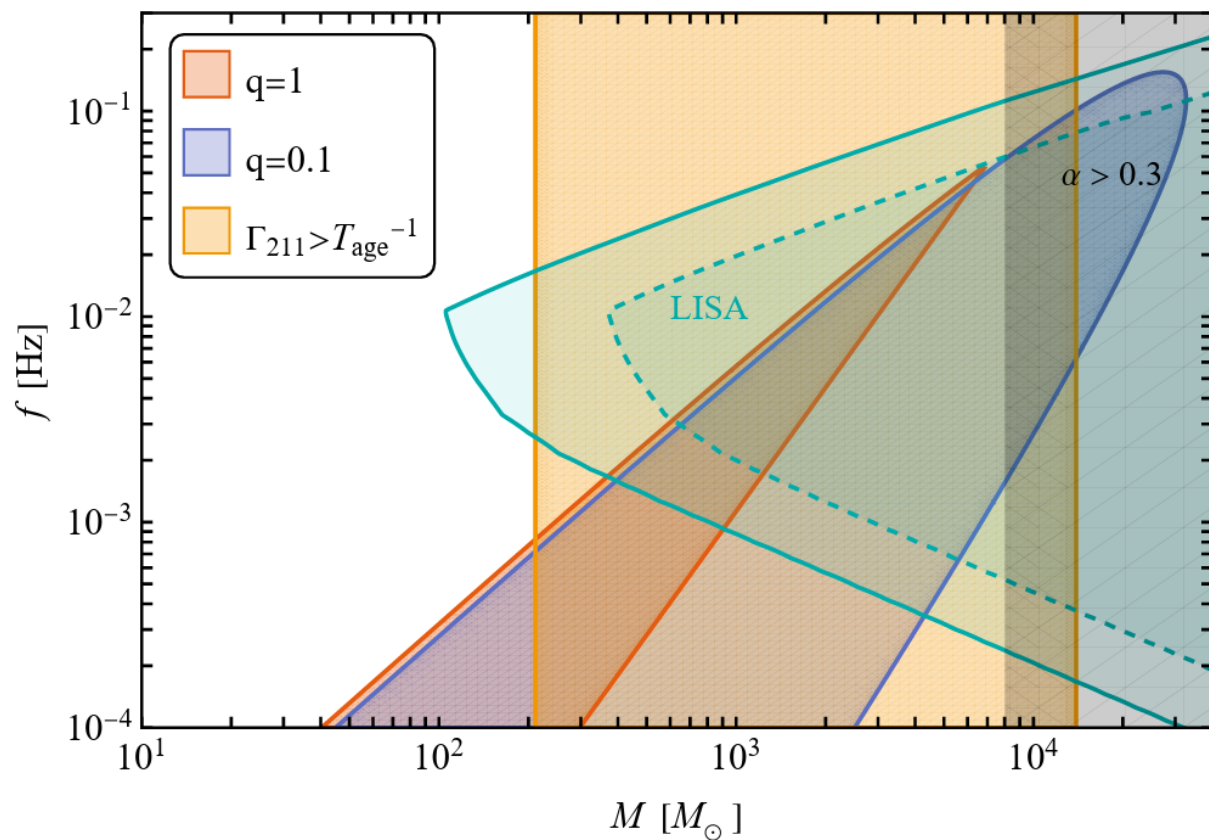
$M = 30 M_{\odot}, q = 1, z = 0.5$

$$P_{\text{DF}} = -4\pi \frac{G^2 M_*^2}{v} \rho_D C_{\Lambda}$$



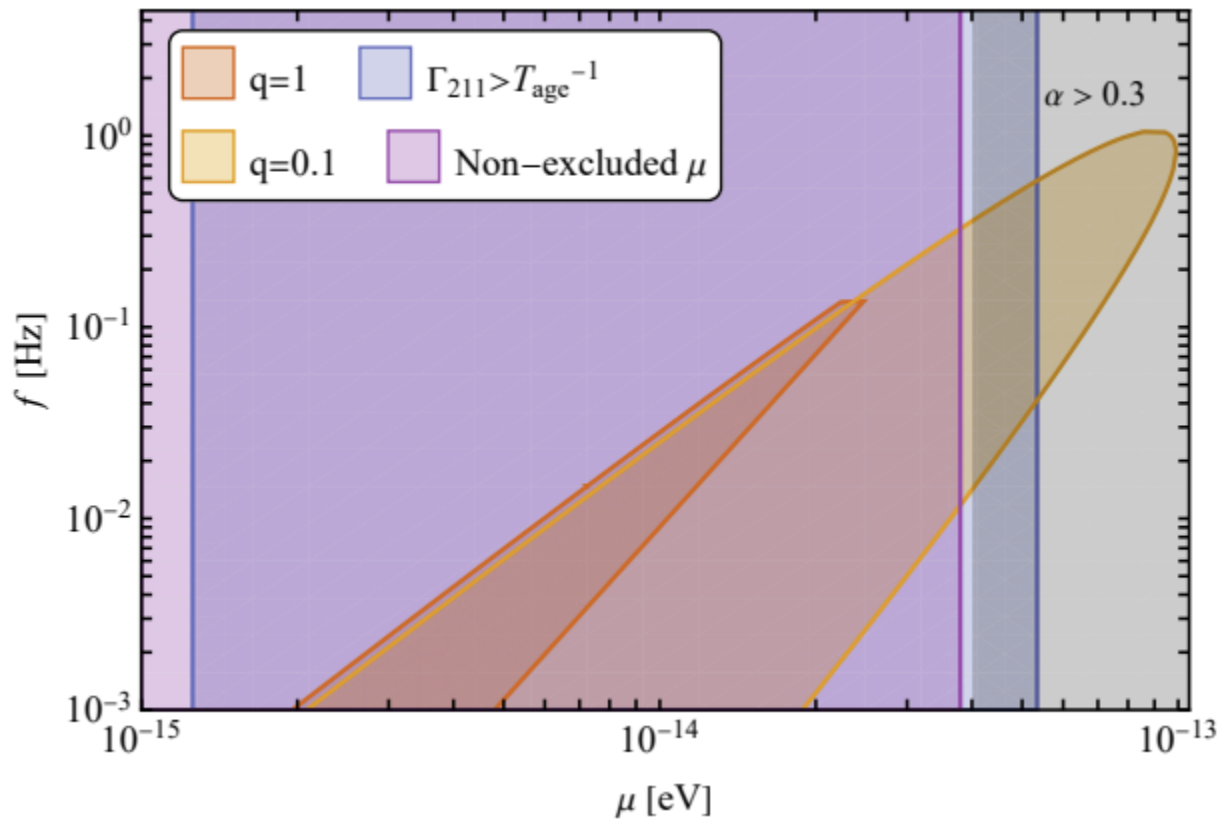
$$\begin{aligned} D &\equiv -h^2 f^3 \frac{dg/dt}{df/dt} \\ &= \frac{12G}{c^4 d_L(z)} |P_{\text{DF}}| \left(\frac{11}{3} - \frac{d \ln \rho}{d \ln f} - \frac{d \ln C_{\Lambda}}{d \ln f} \right) \end{aligned}$$

Gravitational Wave Waveform



$$\mu = 5 \times 10^{-15} \text{ eV},$$

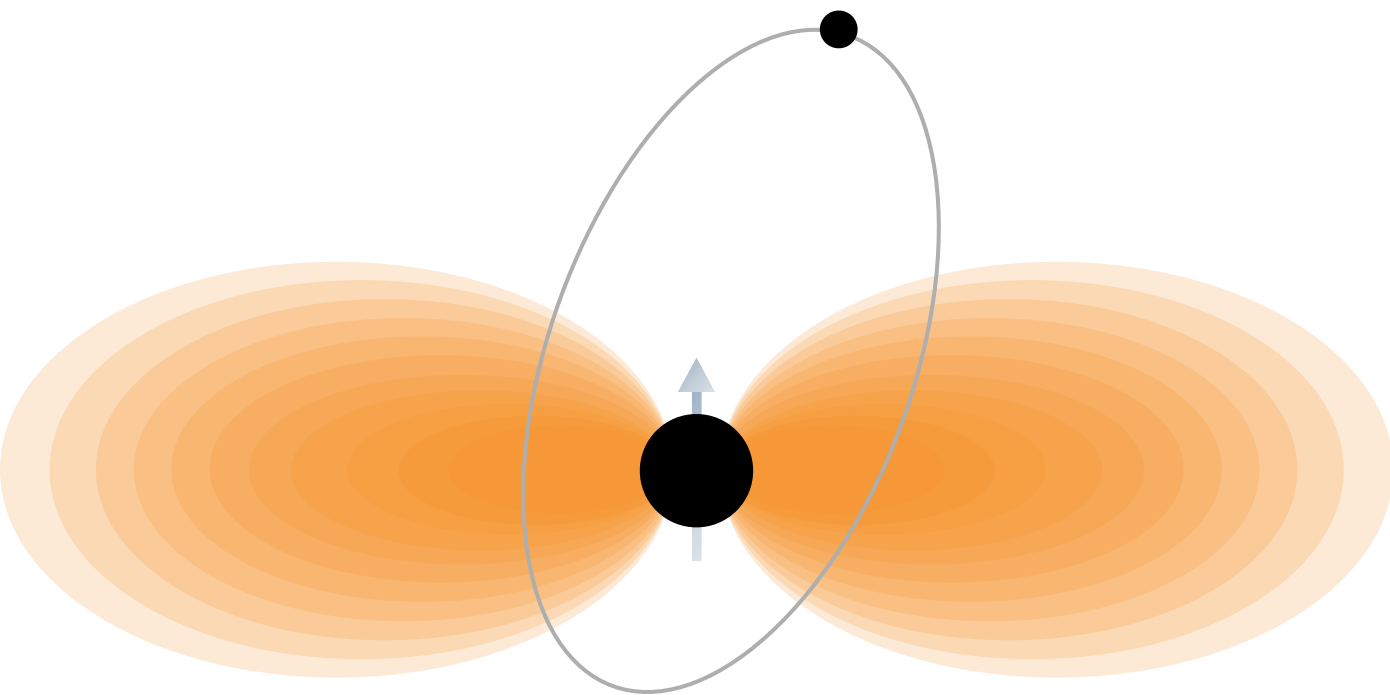
$$z = 0.5$$



$$M_B = 1000 M_\odot,$$

$$z = 0.5$$

Conclusion



Thank you!

Superradiance boson cloud around OJ 287

Reconcile potential tension with

$$\mu = (8.8 - 9.3) \times 10^{-22} \text{ eV}$$

If no such a tension

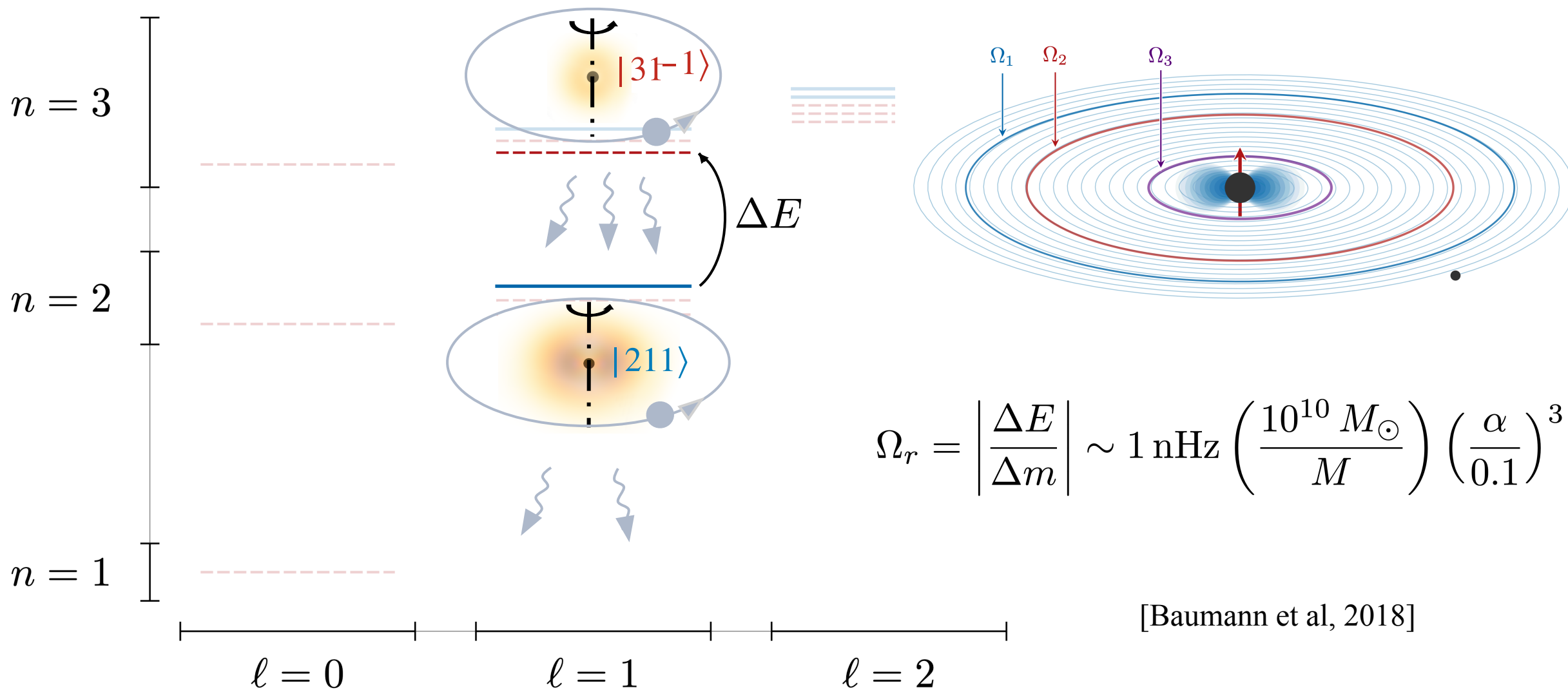
$$\mu \notin (8.5 - 22) \times 10^{-22} \text{ eV}$$

Superradiance boson cloud around SMBH

Partially solve final parsec problem and
leave unique imprint on SGWB and GW
waveform

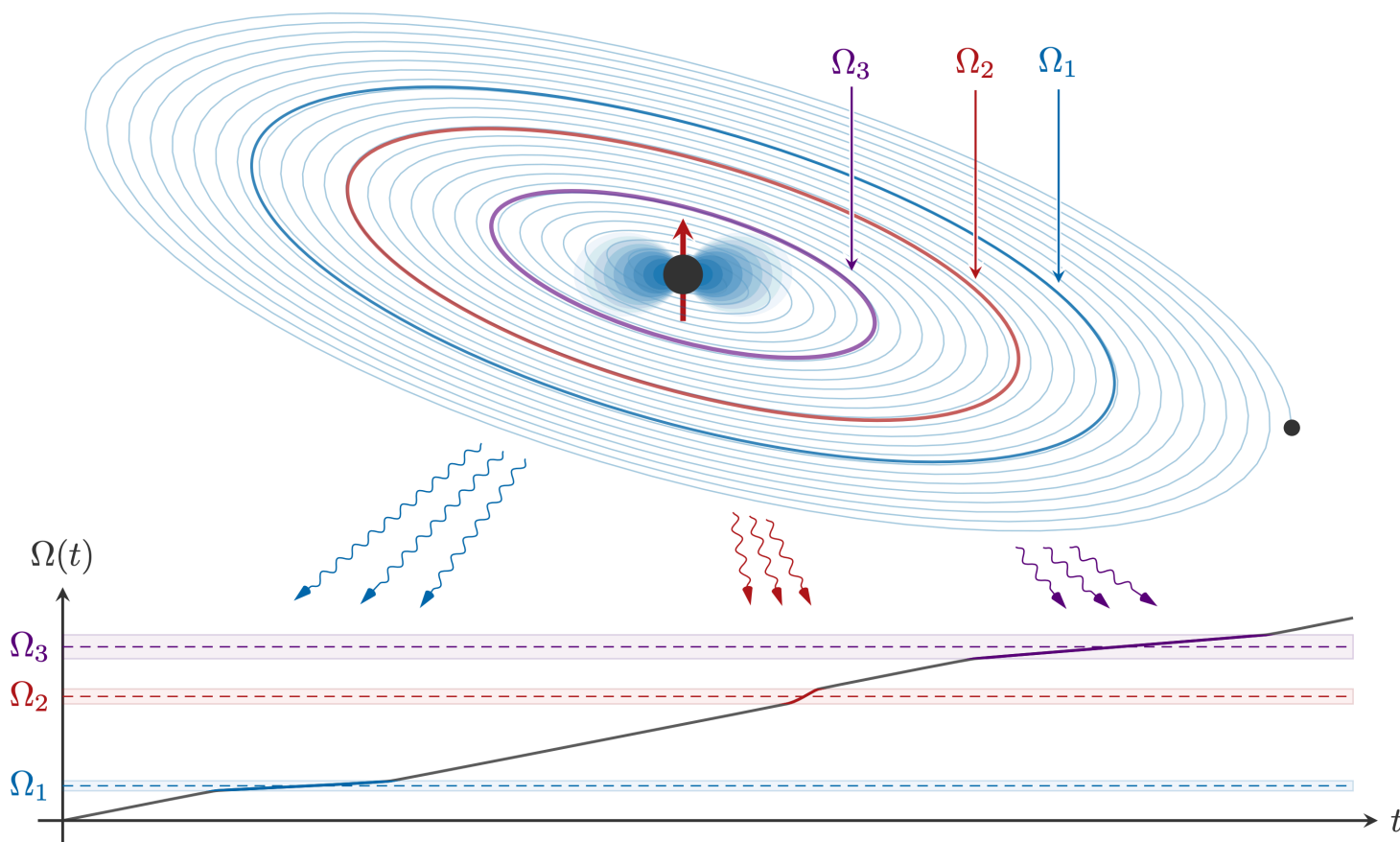
Backup slides

Resonance



[Baumann et al, 2018]

Resonance and Transition Rate



$$R = 1 - e^{-2\pi Z}$$

$$Z \equiv \frac{|\eta|^2}{\gamma}$$

$$\eta(t) = V_{21} \langle 2 | V_* | 1 \rangle \quad \gamma = \frac{d\Omega}{dt}$$

[Baumann et al, 2019]



How about GA with a binary companion?

[Baumann et al, 2019]
 [Baumann et al, 2020]
 [Baumann et al, 2022]

GA phenomenology in binaries

- Atomic resonant transitions $i\partial_t\psi(t, r) = H\psi(t, r)$, with $H = H_0 + V_*(t)$

$$V_* = -\alpha q \sum_{l_* \geq 2} \sum_{|m_*| \leq l_*} \mathcal{E}_{l_* m_*}(\iota_*, \varphi_*) Y_{l_* m_*}(\theta, \phi) \times \left(\frac{r^{l_*}}{R_*^{l_*+1}} \Theta(R_* - r) + \frac{R_*^{l_*}}{r^{l_*+1}} \Theta(r - R_*) \right)$$

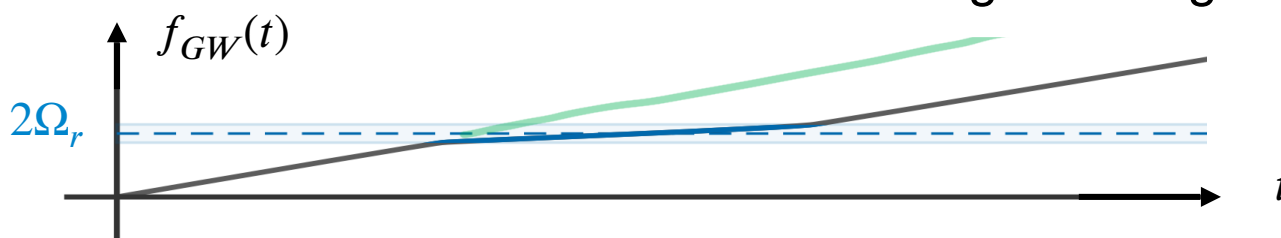
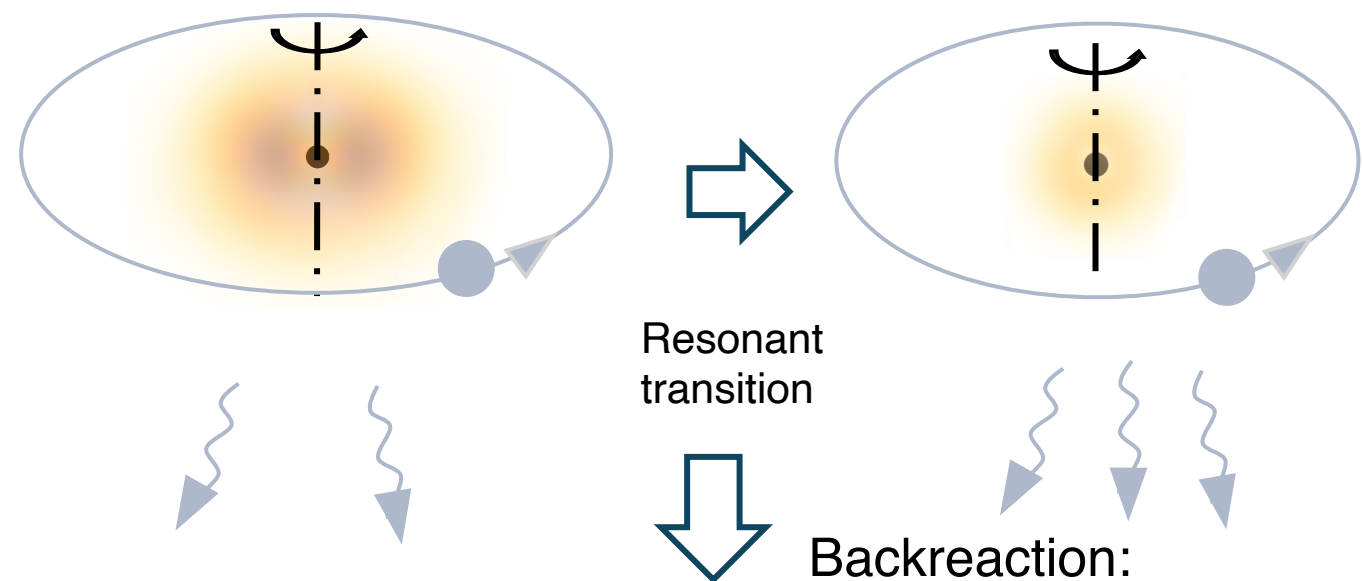
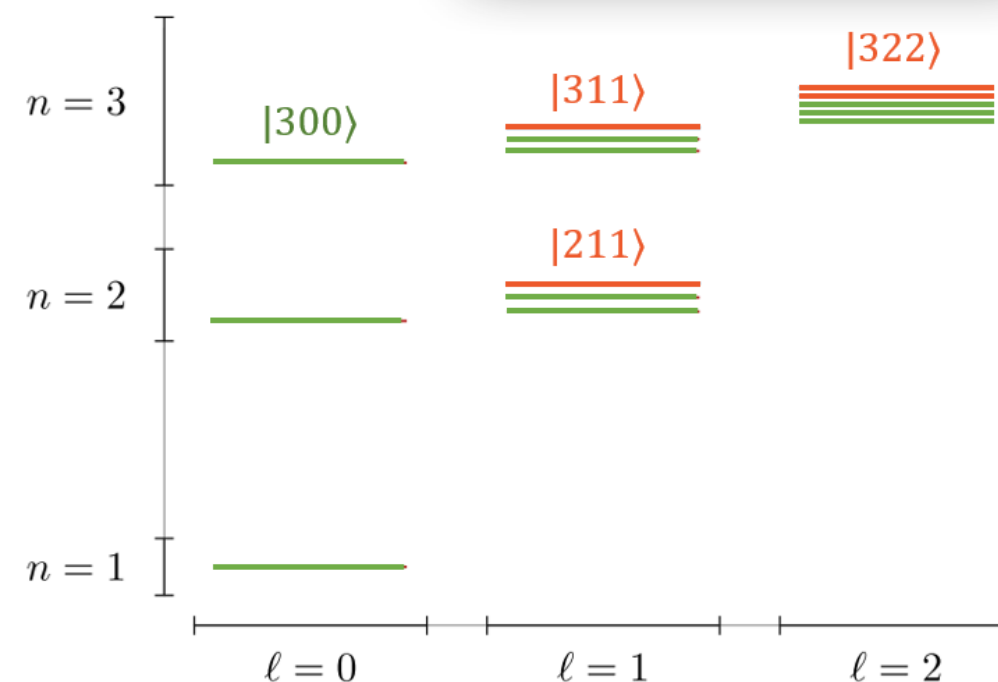
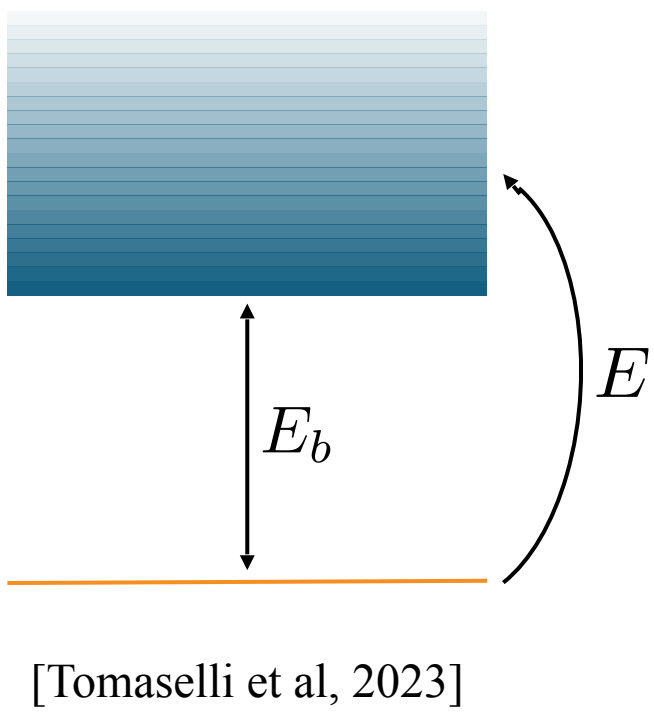
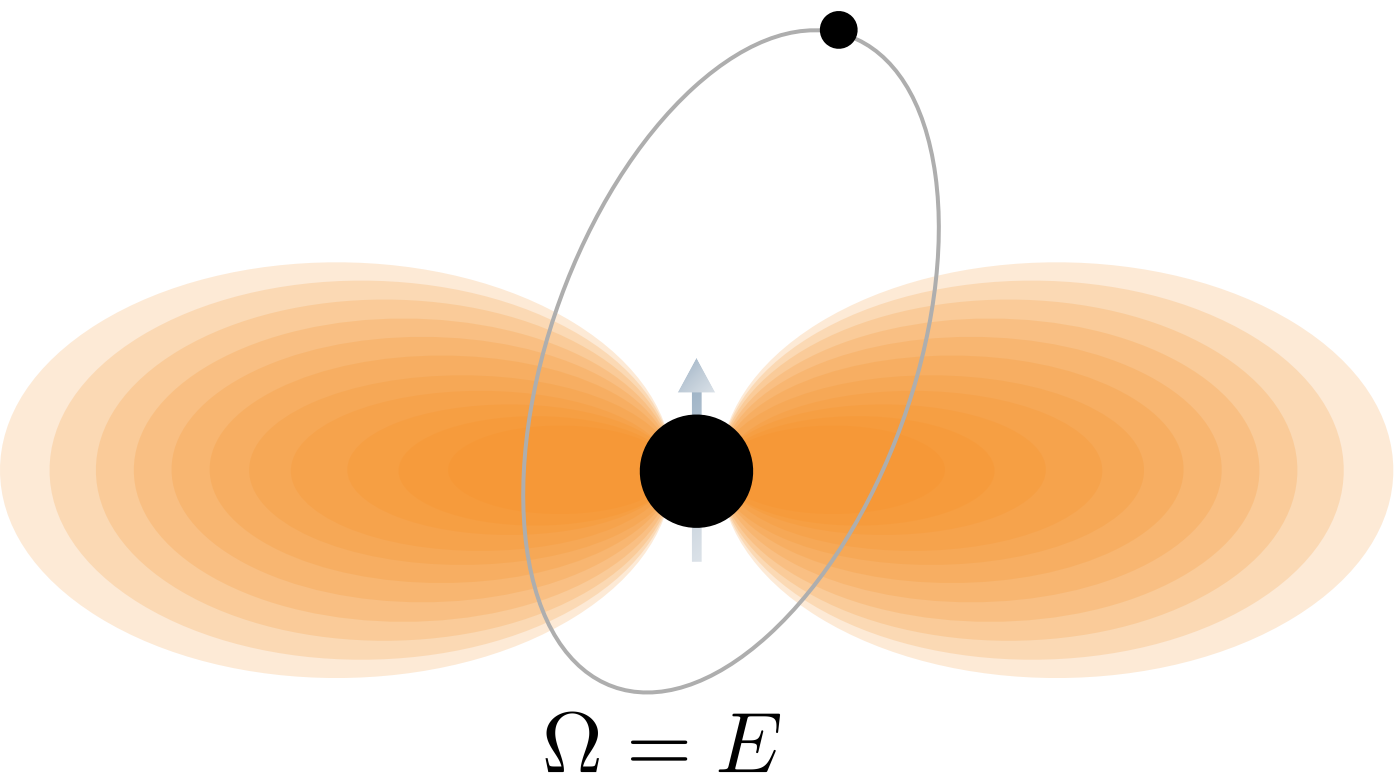


Figure adapted from [Baumann et al, 2020]

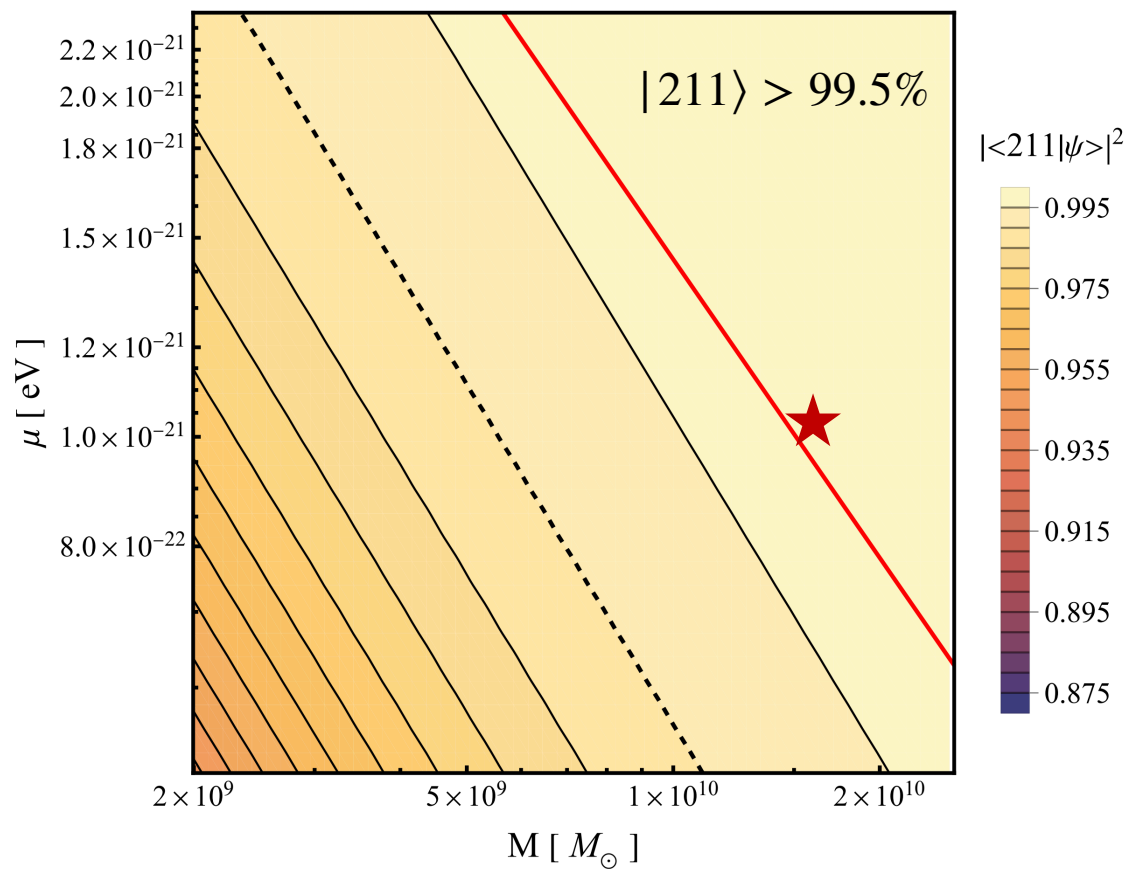


Ionization as Dynamic Friction



$$P_{\text{DF}} = \frac{4\pi q^2 M_{\text{BH}}^2}{v} \rho_{nlm}(R_*, \varphi_*) C_{\Lambda}$$

Cloud State in OJ 287



$$\Gamma_{211} \simeq 1.2 \times 10^{-16} \left(\frac{\alpha}{0.1} \right)^9 \left(\frac{1.8 \times 10^{10} M_{\odot}}{M_{\text{BH}}} \right) \text{s}^{-1}$$

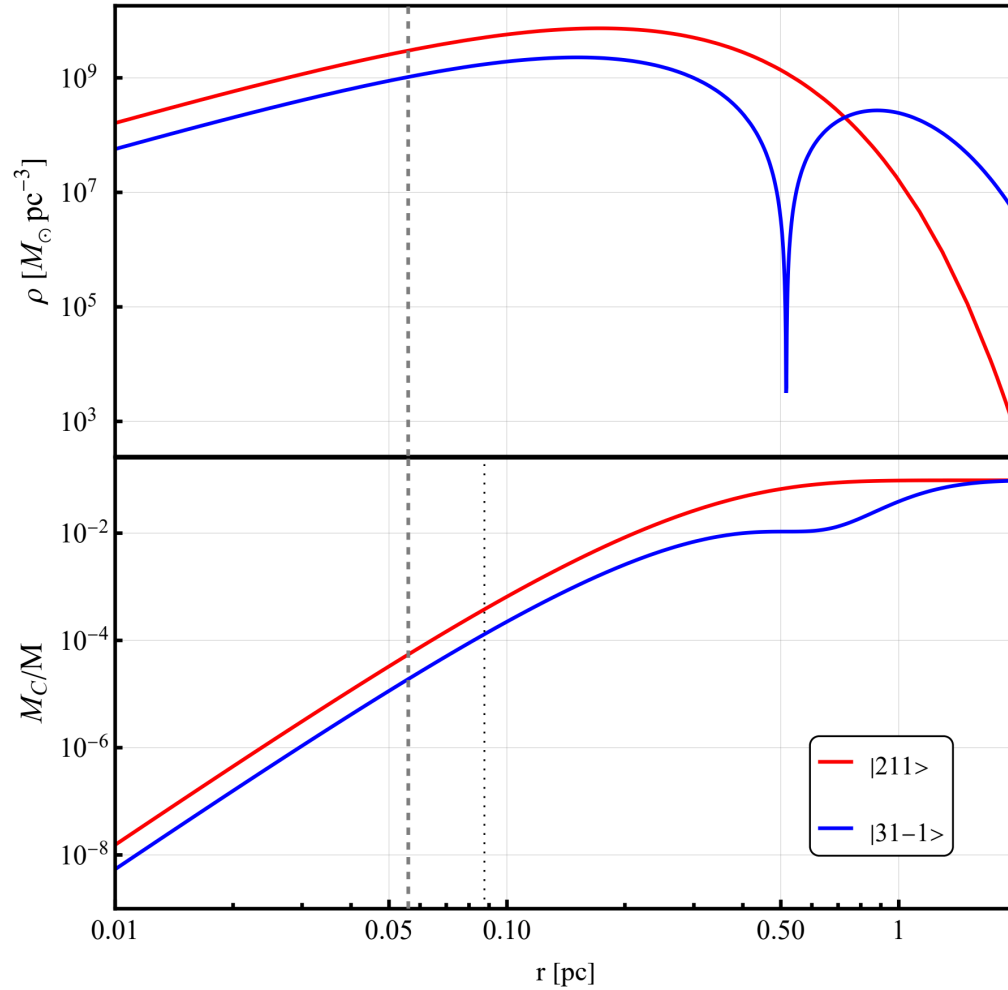
$$\Omega_r = \left| \frac{\Delta E}{\Delta m} \right| \sim 1 \text{ nHz} \left(\frac{10^{10} M_{\odot}}{M} \right) \left(\frac{\alpha}{0.1} \right)^3$$

$$T_{211} \sim \frac{1}{\Gamma_{211}} = 0.0028 T_{\text{age}}$$

$$\Omega_{\text{OJ287}} \simeq 3.5 \text{ nHz} > \Omega_{|211\rangle \rightarrow |31-1\rangle} \simeq 1.3 \text{ nHz}$$

$$R = 1 - e^{-2\pi Z} \quad Z \equiv \frac{|\eta|^2}{\gamma}$$

Cloud Density Profile around OJ 287



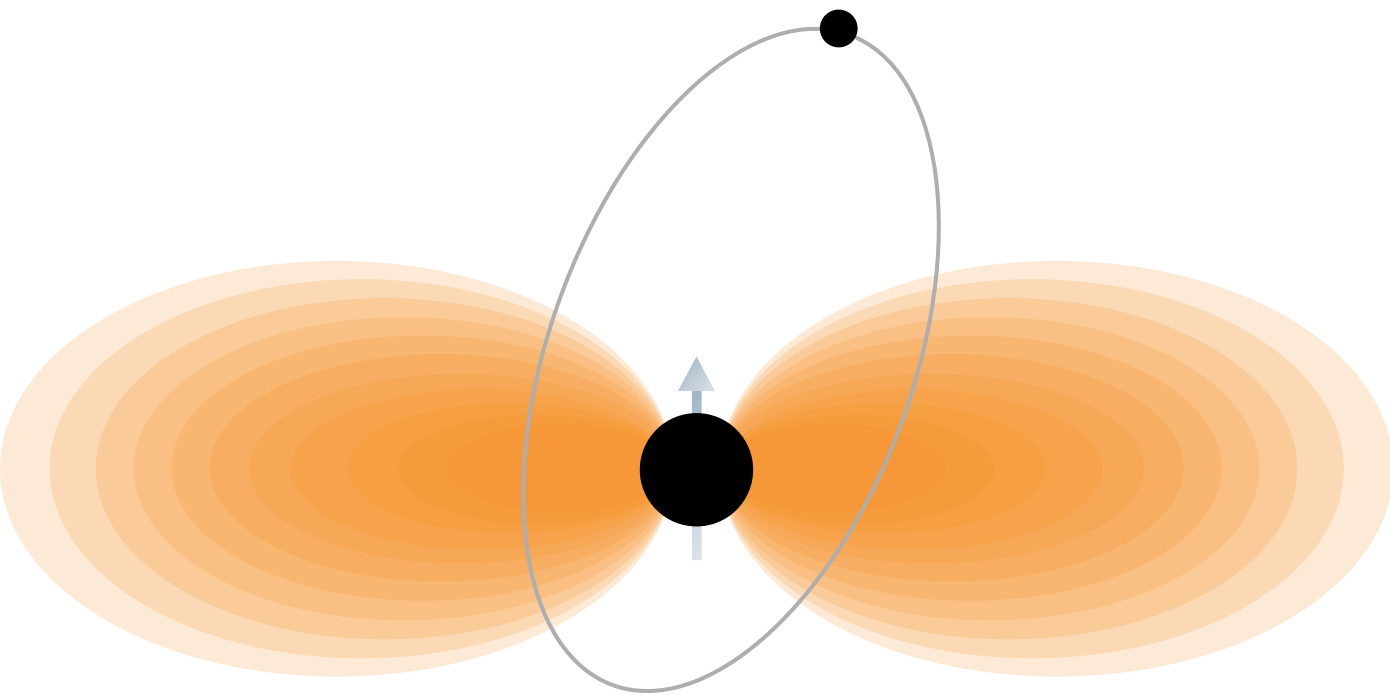
$$\rho_{nlm} = M_{nlm} |\psi_{nlm}|^2$$

$$\psi_{nlm} = R_{nl}(r) Y_{lm}(\theta, \phi) e^{-i(\omega_{nlm} - \mu)t}$$

Upper bound from OJ 287 flare timing

$$\int_{r < 50r_s} \rho_{nlm} dV < 3\%$$

Dynamic Friction Power

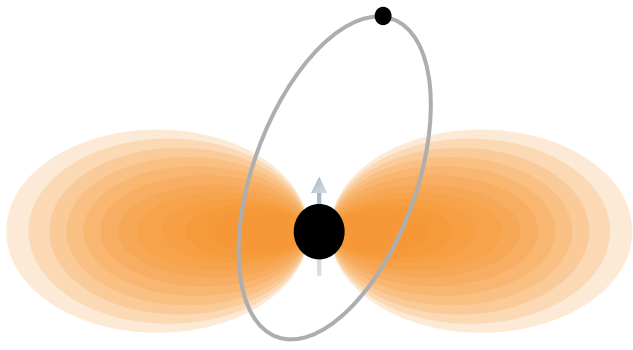


$$P_{\text{DF}} = \frac{4\pi q^2 M_{\text{BH}}^2}{v} \rho_{nlm}(x_*, \theta_*) C_{\Lambda}$$

$$\rho_{211}(x, \theta) = \frac{\beta}{64\pi} \frac{M_{\text{BH}}}{r_0^3} x^2 e^{-x} \sin^2 \theta$$

$$R_*(\varphi_*) = \frac{a(1 - e^2)}{1 + e \cos(1 - \zeta)\varphi_*}$$

$$\langle P_{\text{DF}} \rangle = \frac{1}{T} \int_0^{2\pi} P_{\text{DF}}(R_*(\varphi_*), \theta_*(\varphi_*, \iota)) \frac{dt}{d\varphi_*} d\varphi_*$$

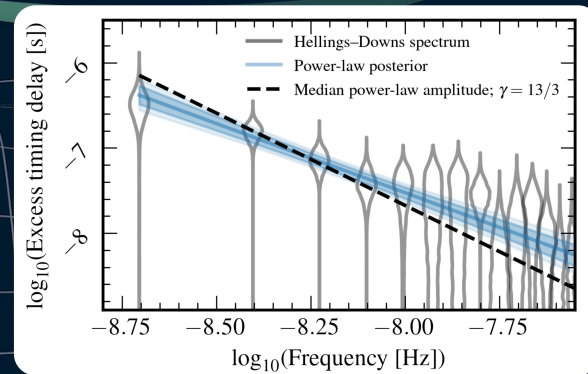
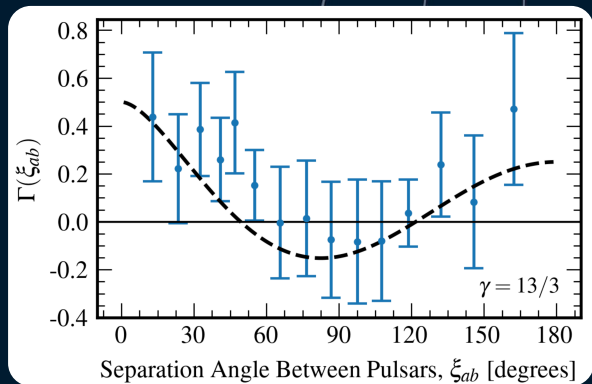


OJ 287 Parameters

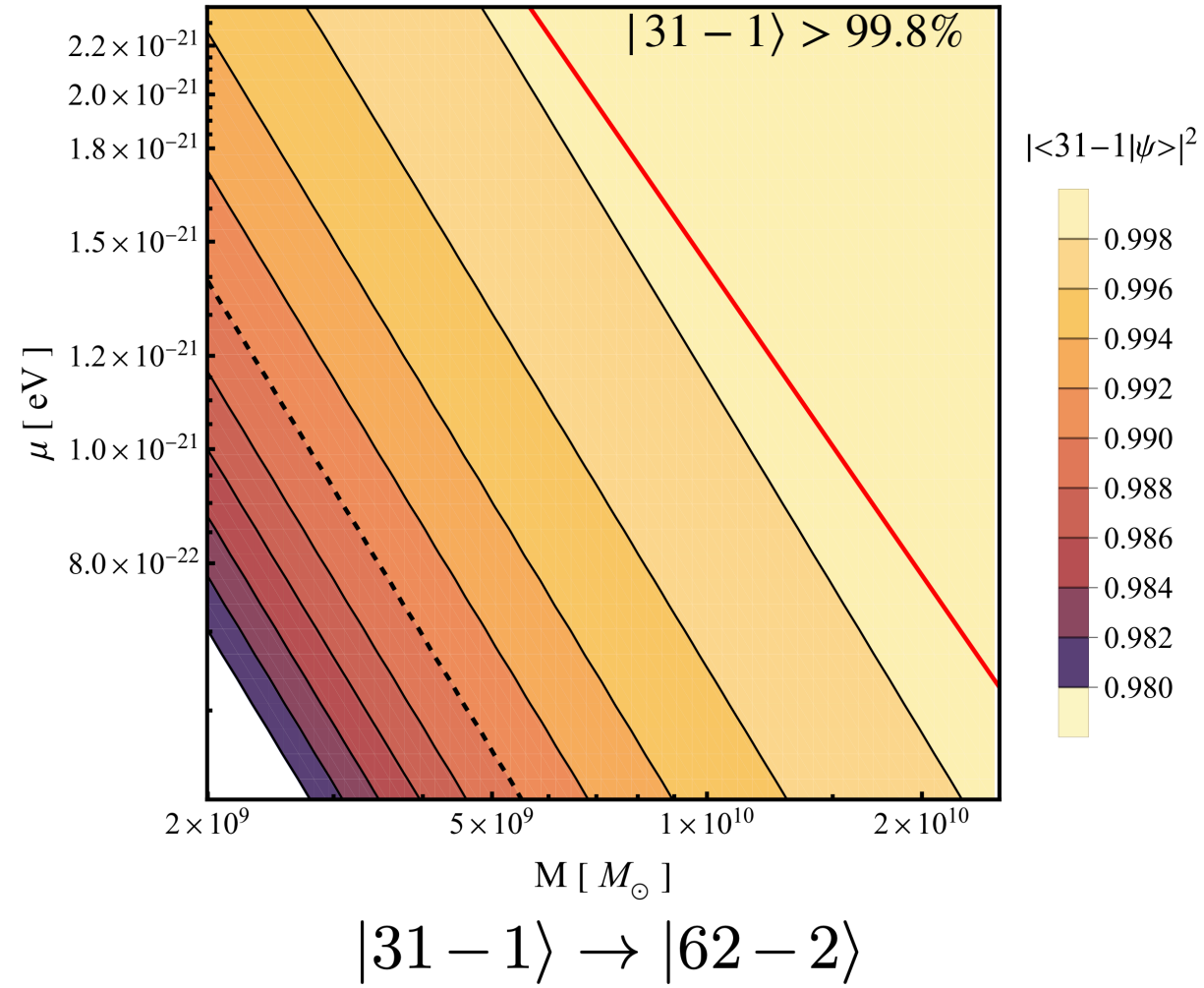
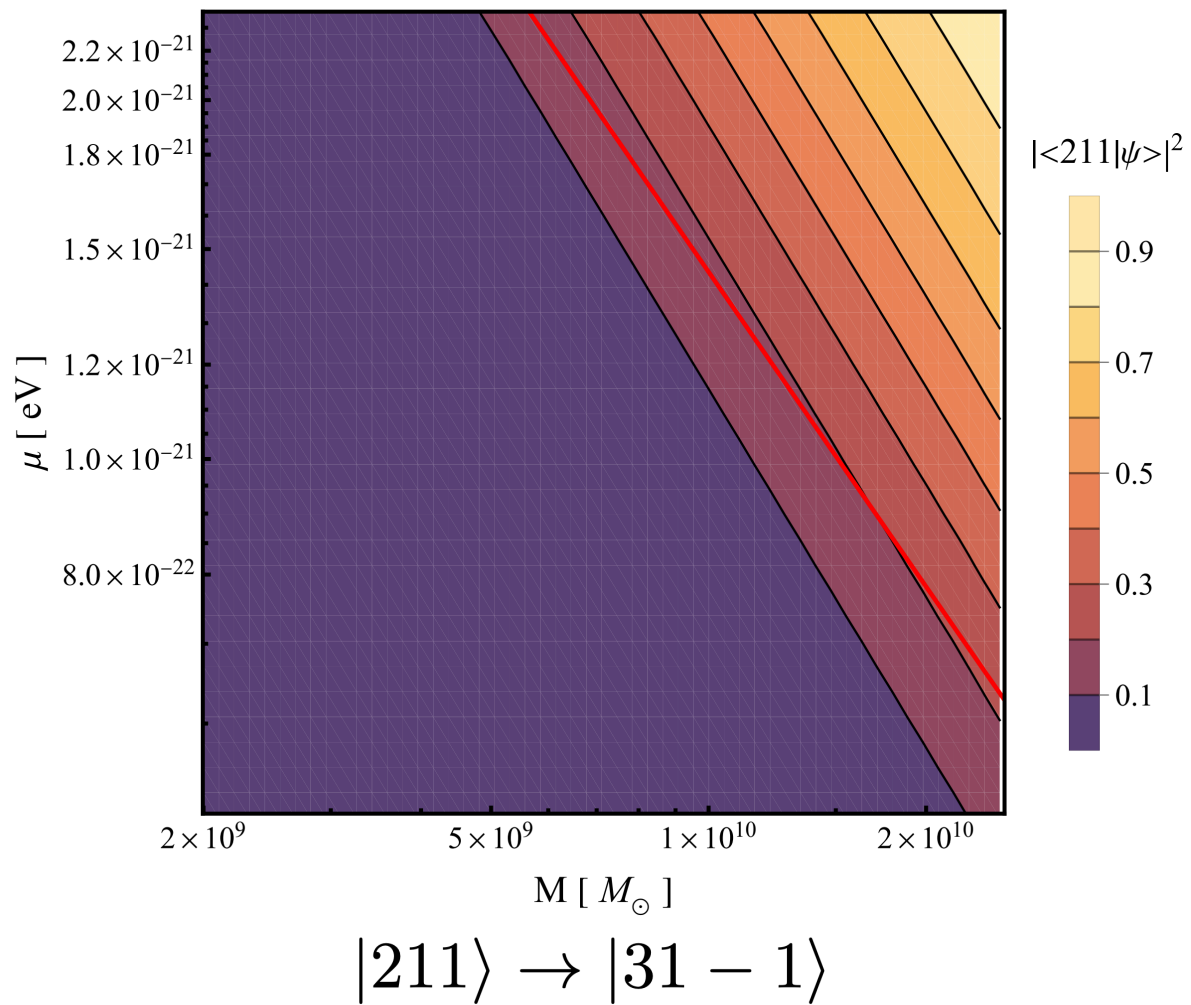
Parameters	Value	Ref.
Primary SMBH mass ($10^{10} M_{\odot}$)	1.8348 ± 0.0008	[41]
Secondary SMBH mass ($10^8 M_{\odot}$)	1.5013 ± 0.0025	[41]
Eccentricity e	0.657 ± 0.001	[41]
Precession angle per period $\Delta\vartheta$	$38.62^{\circ} \pm 0.01^{\circ}$	[41]
Orbital period T (yr)	12.067 ± 0.007	[41]
Orbital period decay rate $\dot{T}$	-0.00099 ± 0.00006	[41]
Spin of the primary SMBH χ_{P_2}	0.381 ± 0.004	[41]
Spin of the primary SMBH χ_{P_1}	0.313 ± 0.01	[54]
Semi-major axis a (pc)	0.056	[42]
Inclination angle of orbit plane	90°	[43]

TABLE S1. Parameters characterizing OJ287 BH system.

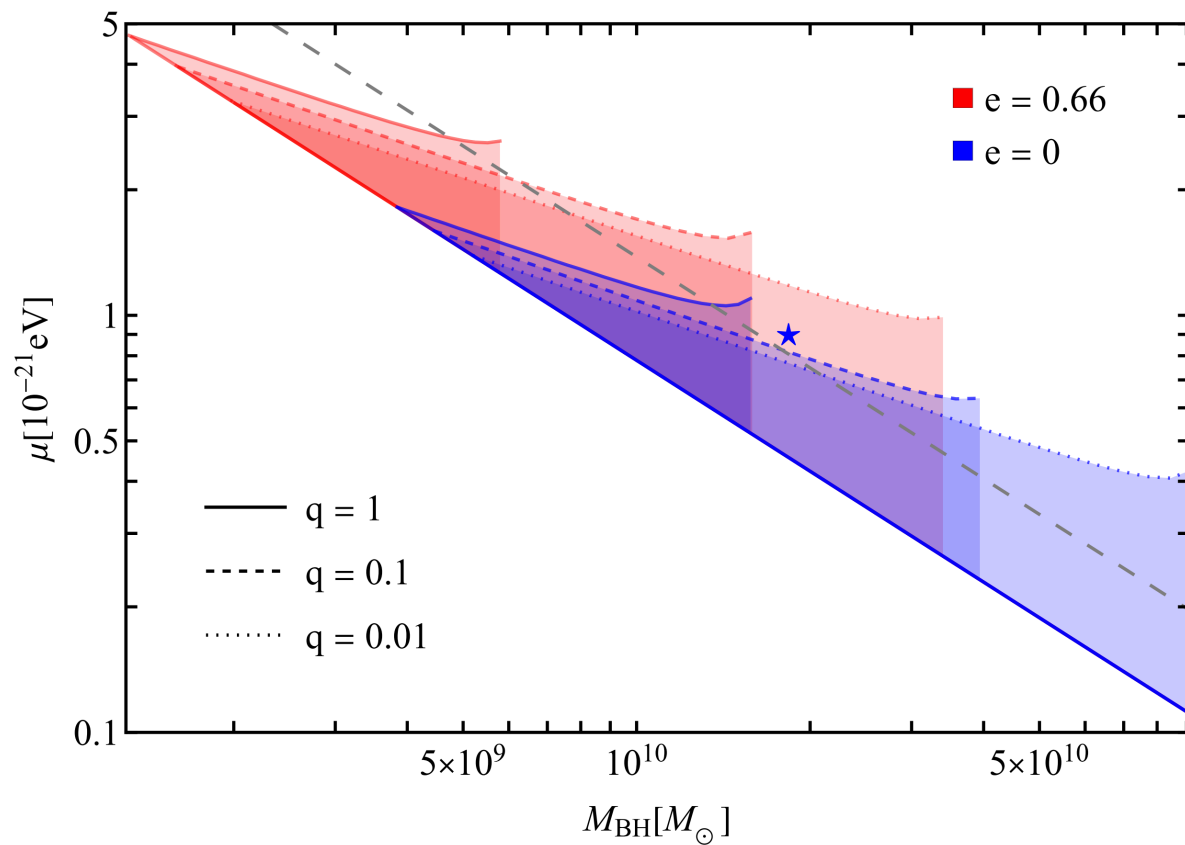
Other Imprints?



Boson Cloud Transition in Final Parsec



Final Parsec Problem



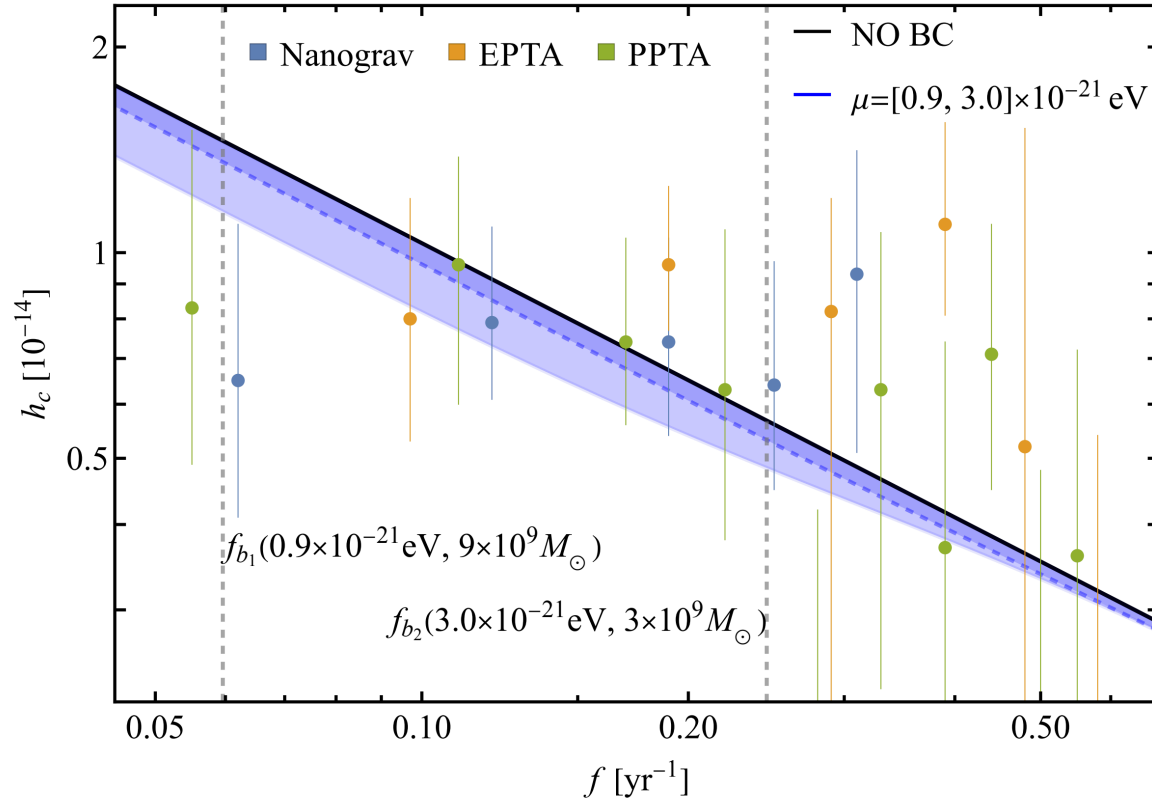
$$\frac{1}{a} \frac{da}{dt} = \frac{2a}{qM_{\text{BH}}^2} (\langle P_{\text{DF}} \rangle + \langle P_{\text{GW}} \rangle)$$

$$P_{\text{DF}} = \frac{4\pi q^2 M_{\text{BH}}^2}{v} \rho_{nlm}(x_*, \theta_*) C_{\Lambda}$$

$$\rho = \eta \rho_{211} + (1 - \eta) \rho_{31-1}$$

$$\int_{0.01 \text{ pc}}^{\mathcal{O}(1) \text{ pc}} \frac{dt}{da} da < 1 \text{ Gyr}$$

Stochastic Gravitational Wave Background



$$\frac{dE_{\text{GW}}}{df_s} = \frac{\pi^{2/3}}{3} \frac{M_{\text{BH}}^{5/3}}{f_s^{1/3}} \frac{q}{(1+q)^{1/3}} \frac{\langle P_{\text{GW}} \rangle}{\langle P_{\text{GW}} \rangle + \langle P_{\text{DF}} \rangle}$$

$$h_c^2(f) = \frac{4}{\pi f} \int dz dq dM_{\text{BH}} \frac{d^3 n}{dz dq dM_{\text{BH}}} \frac{dE_{\text{GW}}}{df_s}$$

$$f_b \simeq 3.2 \text{ nHz} \left(\frac{\mu}{10^{-21} \text{ eV}} \right)^3 \left(\frac{M_{\text{BH}}}{10^{10} M_\odot} \right)^2$$